POST AND TELECOMMUICATIONS INSTITUTE OF TECHNOLOGY

Interfacing with memory and input/output devices

Faculty: Computer science – IT1

Instructor: Viet Hoang Pham

Email: vietph@ptit.edu.vn

Phone: 0945296388

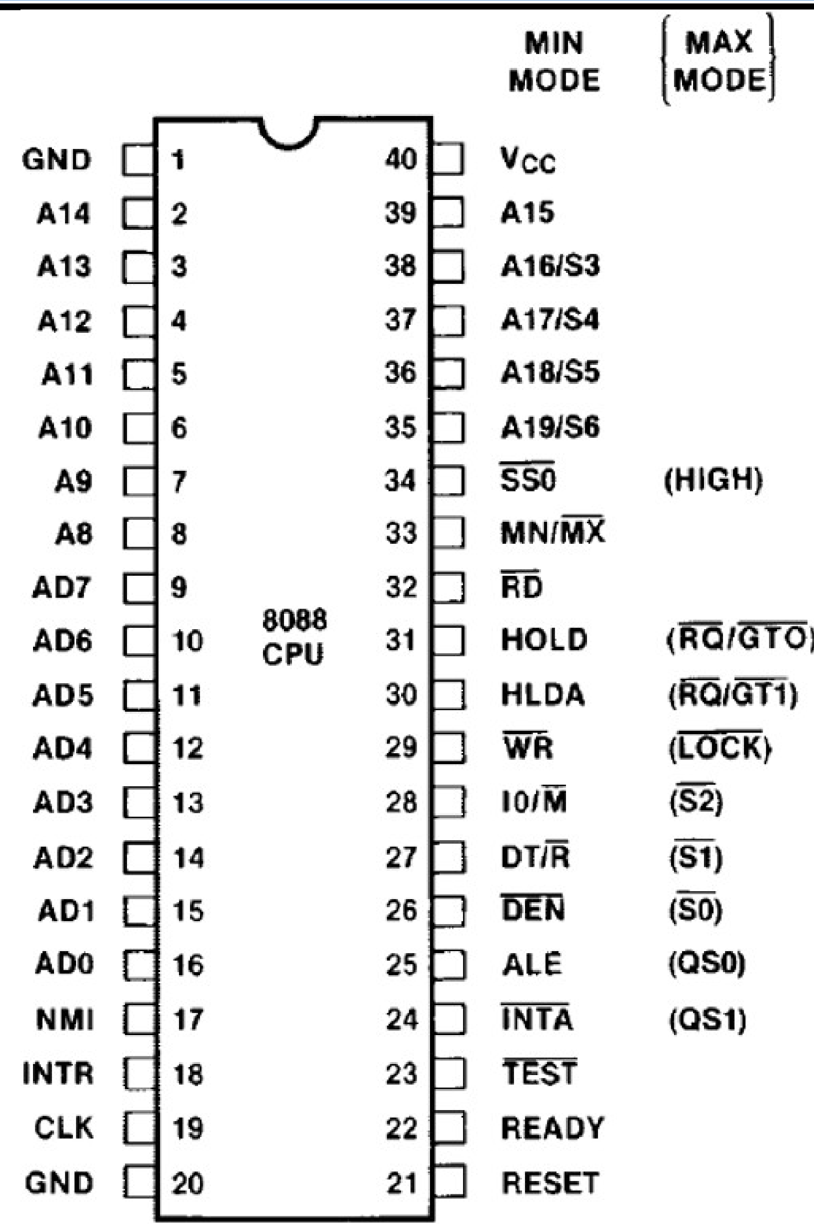
Contents

- CPU signals
- Auxiliary circuits
- CPU-memory interface
- Introduction to memory
  - Memory address decoding
  - Memory address decoding using basic logic circuits
  - Memory address decoding using integrated circuits
  - Memory address decoding using EPROM memory chips
- CPU-input/output device interface
- Some simple gate circuits.

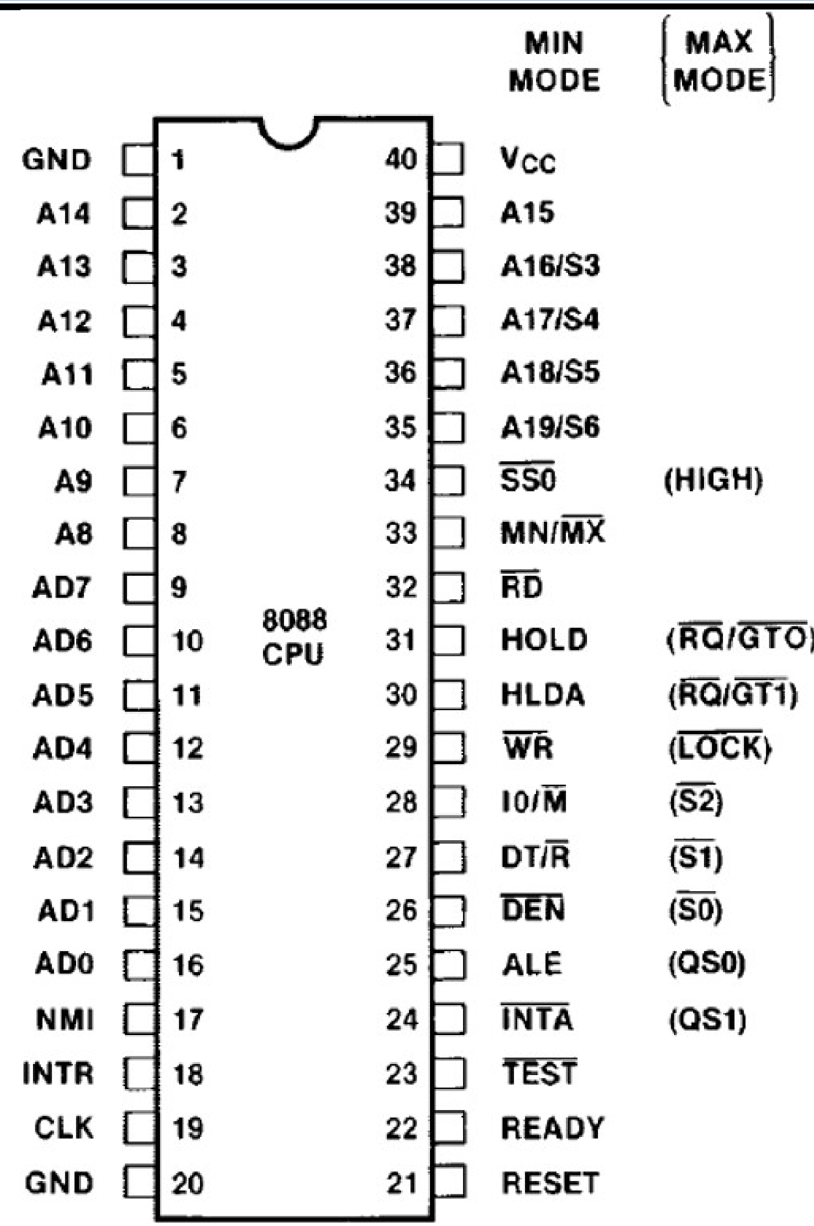
CPU signals

❖ The 8088 microcontroller has 40 signal pins:

- Address signal group:
  - AD0-AD7: 8 multiplexer pins for the low end of bus A and bus D;
  - A8-A15: 8 signal pins for the high end of bus A;
  - A16/S3-A19/S6: 4 multiplexer pins for the high end of bus A and bus C;



- The 8088 microcontroller has 40 signal pins:
  - Data signal group:
    - AD0-AD7: 8 multiplexer pins for the low end of bus A and bus D;
    - When the latch pin ALE=0, it is a data signal; when ALE=1, it is an address signal.



CPU signals

❖ The 8088 microcontroller has 40 signal pins:

- System control signal group:
  - IO/M: CPU selects whether to work with an input/output device or memory.
    - –IO/M = 1: CPU selects to work with an input/output device.
    - –IO/M = 0: CPU selects to work with memory. The corresponding address of the selected component appears on the address bus.

CPU signals

❖ The 8088 microcontroller has 40 signal pins:

- System control signal group:
  - DT/R: Signal determining the direction of data transport on the data bus.
    - –DT/R = 1: Data goes out from the CPU;
    - –DT/R = 0: Data goes to the CPU.
  - RD: Read enable (reverse) pulse. When RD = 0, the data bus is ready to receive data from memory or peripheral devices.
  - WR: Write enable signal. When WR = 0, the data is stable on the data bus and is written to memory or an input/output device when WR = 1.

CPU signals

❖ The 8088 microcontroller has 40 signal pins:

- System control signal group:
  - DEN: A signal that informs the external circuit that the data has stabilized on the data bus.
  - SS0: A status signal used in conjunction with IO/M and DT/R to decode bus operation cycles.
  - READY: A signal that informs the CPU of the readiness status of the peripheral device or memory.
    - When READY = 1, the CPU can perform read/write operations immediately without inserting additional wait cycles;
    - When the peripheral device or memory is not ready, it sends READY=0 to the CPU to extend the read/write command by adding wait cycles.

CPU signals

- The 8088 microcontroller has 40 signal pins:
  - Bus control signal group - HOLD: Signal requesting the CPU to suspend so that the external circuit can exchange data with memory using direct memory access.
    - When HOLD=1, the CPU will suspend itself by disconnecting from buses A, D, and part of bus C, allowing the DMAC circuit to control the direct data exchange process between memory and input/output devices.
    - HLDA: Signal informing the external circuit that the CPU suspend request has been accepted. The CPU suspends by disconnecting from buses A, D, and some signals of bus C.

CPU signals

- The 8088 microcontroller has 40 signal pins:
  - Bus control signal group - HOLD: Signal requesting the CPU to suspend so that the external circuit can exchange data with memory using direct memory access.
    - INTA: Signal informing the external circuit that the INTR interrupt request has been accepted. The CPU outputs INTA=0 to inform the external circuit that it is waiting for the external circuit to place the interrupt number on the data bus.
    - ALE: Address latch pulse determines whether the signal on the AD multiplexing pins is an address or data signal. When ALE=1, the signal on the AD multiplexing pins is an address signal.

CPU signals

- The 8088 microcontroller has 40 signal pins:
  - CPU Control Signal Group:
    - NMI: Unmasked interrupt request signal – not restricted by the IF interrupt flag. Upon receiving an NMI interrupt request, the CPU completes the current instruction and enters the interrupt service cycle.
    - INTR: Masked interrupt request signal – restricted by the IF interrupt flag.
      - An INTR interrupt request will be denied when the IF interrupt flag is 0.
      - Upon receiving an INTR interrupt request and the IF interrupt flag is 1, the CPU completes the current instruction and enters the interrupt service cycle, sending an interrupt acknowledgment signal INTA=0.

CPU signals

- The 8088 microcontroller has 40 signal pins:
  - CPU Control Signal Group:
    - RESET: 8086/8088 restart signal. When RESET = 1 persists for at least 4 clock cycles, the 8086/8088 is forced to restart: it clears the DS, ES, SS, IP, and FR registers to 0 and begins executing the program at address CS:IP=FFFF:0000H.
    - MN/MX: This pin determines the CPU's operating mode (MIN or MAX).
      - In MIN mode (MN/MX connected to 5V), the CPU generates bus control signals;
      - In MAX mode (MN/MX connected to ground), the CPU transmits status signals to an external circuit to generate bus control signals.
    - TEST: The TEST signal is checked by the WAIT instruction. When the CPU executes the WAIT instruction while TEST = 1, it waits until TEST = 0 before executing the next instruction.

CPU signals

- The 8088 microcontroller has 40 signal pins:
  - Clock and Power Signal Group:
    - CLK: Clock signal providing the CPU's operating clock.
    - Vcc: 5V power supply pin.
    - GND: Ground pin.
  - Status Signal Group:
    - S3, S4: In combination indicate the access status of segment registers:
      - 00: CPU accesses auxiliary data segment ES
      - 01: CPU accesses stack segment SS
      - 10: CPU accesses code segment or no segment
      - 11: CPU accesses data segment
    - S5: S5 reflects the value of the IF flag
    - S6: S6 is always 0

CPU signals

❖ Bus cycles:

IO/M	DT/R	SS0	Ý nghĩa
0	0	0	Đọc mã lệnh
0	0	1	Đọc bộ nhớ
0	1	0	Ghi bộ nhớ
0	1	1	Buýt rỗi
1	0	0	Chấp nhận yêu cầu ngắt
1	0	1	Đọc thiết bị ngoại vi
1	1	0	Ghi thiết bị ngoại vi
1	1	1	Dừng

CPU signals

❖ The microcontroller can operate in two modes: Min and Max.

- Min Mode:
  - MN/MX pins are connected to 5V power supply.
  - The CPU generates control signals for memory and traditional peripherals.
  - Signals: IO/M, WR, INTA, ALE, HOLD, HLDA, DT/R, DEN.
- Max Mode:
  - MN/MX pins are connected to ground.
  - The CPU sends status signals to auxiliary circuits, and these circuits generate control signals for memory and peripherals.
  - Signals: RQ/GT0, RQ/GT1, LOCK, S2, S1, S0, QS0, QS1.

CPU signals

❖Max Mode Specific Signals

- RQ/GT0 and RQ/GT1: These signals request the use of the bus by other processors or indicate that the CPU accepts a hang to allow other processors to use the bus. RQ/GT0 has higher priority than RQ/GT1.
- LOCK: This signal is given by the CPU to prohibit other processors in the system from using the bus while it is executing an instruction with a LOCK prefix.
- QS0, QS1: These signals indicate different states of the instruction buffer (instruction queue).
- S2, S1, and S0: Status pins used in MAX mode to interface with the 8288 bus controller. These signals are used by the 8288 to generate control signals during bus cycles.

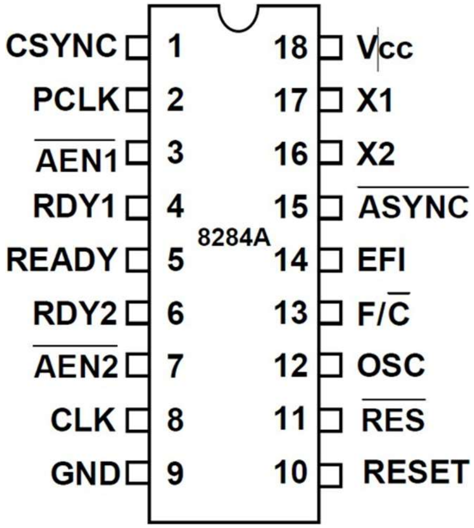
❖Max Mode Specific Signals

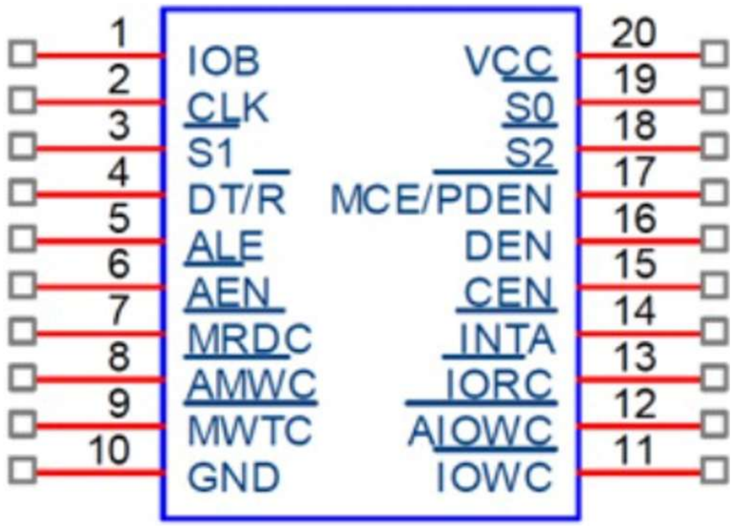
$\overline{S2}$	$\overline{S1}$	$\overline{S0}$	Chu kỳ điều khiển của buýt	Tín hiệu
0	0	0	Chấp nhận yêu cầu ngắt	INTA
0	0	1	Đọc thiết bị ngoại vi	IORC
0	1	0	Ghi thiết bị ngoại vi	IOWC, $\overline{AIOWC}$
0	1	1	Dừng (halt)	Không
1	0	0	Đọc mã lệnh	MRDC
1	0	1	Đọc bộ nhớ	MRDC
1	1	0	Ghi bộ nhớ	MWTC, $\overline{AMWC}$
1	1	1	Buýt rỗi (nghỉ)	Không

Auxiliary circuits

❖ Auxiliary circuits are circuits that provide input signals or support the CPU's control in maximum mode.Typical auxiliary circuits include:

- 8284 clock generator circuit		8288 bus control circuit





Auxiliary circuits

❖ Auxiliary circuit - 8284 clock generation: Provides CLOCK, READY, and RESET signals coupled to the CPU.

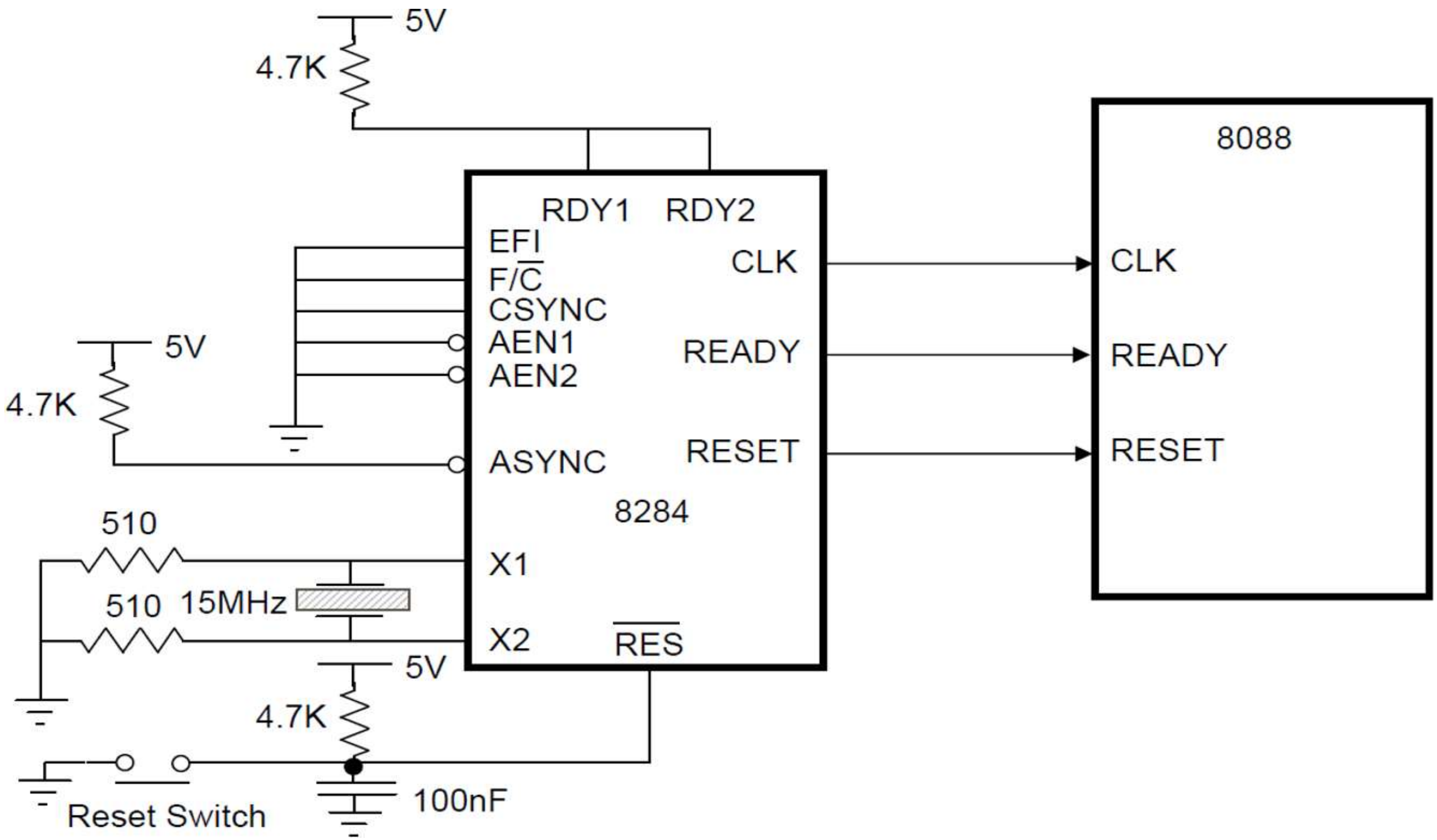
- OSC: Amplified clock signal with frequency equal to fx of the oscillator.
- EFI: External clock input
- CLK: Clock signal (fCLK = fx/3)
- PCLK: External clock signal (FPCLK = fx/6)
- X1, X2: Connected to two pins of the crystal with frequency fx; this crystal is part of an internal oscillator circuit in the 8284 that generates a reference clock signal for the entire system.

❖ Auxiliary circuit - 8284 clock generation:

- AEN1, AEN2: Input select signals corresponding to RDY1, RDY2 indicating the readiness status of memory or peripheral devices.
- RDY1, RDY2: along with AEN1, AEN2 to create wait cycles at the CPU.
- F/C: to select the standard signal source for the 8284. When this pin is high, the external clock signal will be used as the clock signal for the 8284; otherwise, the clock signal of the internal oscillator circuit using the selected crystal will be used as the clock signal.

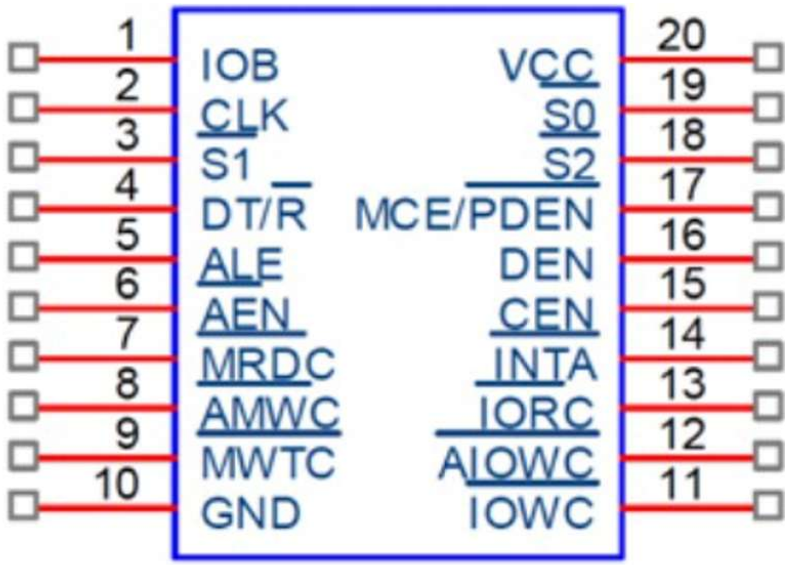
Auxiliary circuits

❖8284 clock generation circuit connected to the CPU:



❖ Auxiliary Circuit - 8288 Bus Control:

- The 8288 bus control circuit receives status signals (S2, S1, and S0) from the CPU and generates bus control signals on behalf of the CPU.
- The 8288 is only used in MAX mode.



Auxiliary circuits

❖ Auxiliary Circuit - 8288 Bus Control:

- Signal pins:
  - S2, S1, and S0: Status input pins from the CPU.
  - CLK: Clock signal taken from the 8284 clock generator circuit to set the operating clock and synchronize with the CPU.
  - CEN: Input signal to enable the DEN signal and other control signals of the 8288.
  - IOB: Signal to control the 8288 circuit to operate in different bus modes.
    - When IOB = 1, the 8288 operates in input/output bus mode;
    - when IOB = 0, the 8288 circuit operates in system bus mode.

❖ Auxiliary Circuit - 8288 Bus Control:

- Signal pins:
  - MRDC: Memory read control signal. It activates the memory to load data onto the bus.
  - MWTC, AMWC: Memory write control signals or extended memory write. AMWC (advanced memory write command) is similar to MWTC, but increases the write time for slow-speed memories.
  - IORC: Peripheral device read control signal - activates selected devices to load data onto the bus.
  - IOWC, AIOWC: Peripheral device write control signals or extended peripheral device write. AIOWC (advanced IO write command) is similar to IOWC, but increases the write time for slow-speed peripheral devices.

❖ Auxiliary Circuit - 8288 Bus Control:

- Signal pins:
  - DT/R: Signal that determines the direction of data transport on the data bus.
    - –DT/R=1 → data goes out from the CPU;
    - –DT/R=0 → data goes to the CPU.
  - DEN: This signal controls whether the data bus becomes a local bus or a system bus.

Auxiliary circuits

❖ Auxiliary Circuit - 8288 Bus Control:

- Signal pins:
  - INTA: Signal to the external circuit that the INTR interrupt request has been accepted. The CPU outputs INTA=0 to inform the external circuit that it is waiting for the external circuit to load the interrupt number onto the data bus.
  - ALE: Address latch pulse. This determines whether the signal on the AD multiplexing pins is an address or data signal. When ALE=1, the signal on the AD multiplexing pins is an address signal.
  - MCE/PDEN: This signal is used to set the operating mode for the PIC 8259 interrupt controller so that it operates in master mode.Send feedback

❖Bus Timing and Read/Write Cycle

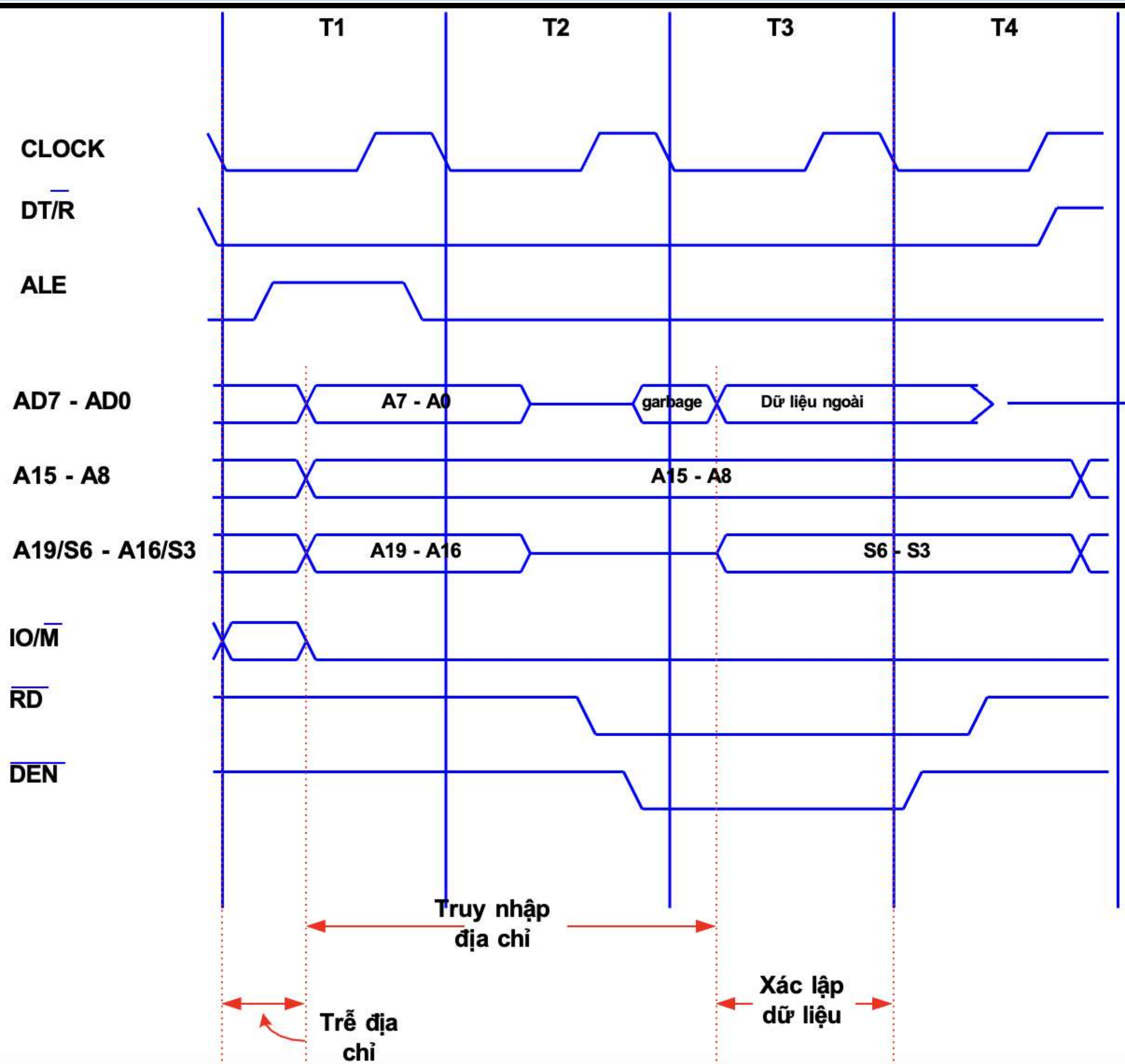
- Memory access, input/output is calculated according to the bus cycle. A typical bus cycle consists of 4 clock pulses (T):
  - Generate address signal on the address bus (T1)
  - Generate read/write signals in the pulse (T2 - T3)
  - Read/Save data on the data bus (T3)
- To transmit data without errors, the signals on the bus need to be generated and maintained during the bus cycle:
  - Impedance distortion (inductance, capacitance)
  - Signal delay during propagation on the bus
  - Pulse shape (rising edge, falling edge, width)

❖Bus Timing and Read/Write Cycle

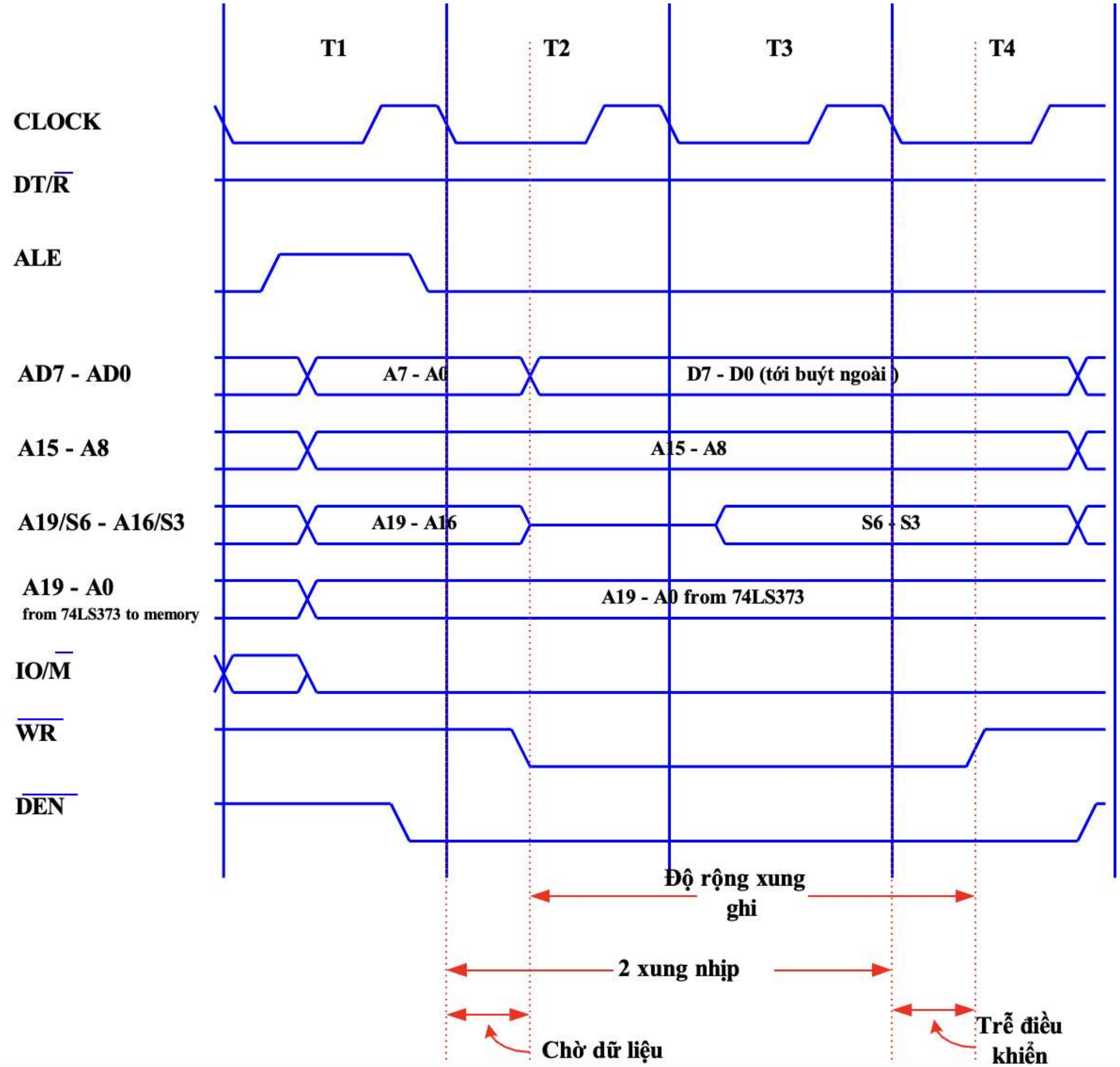
- 4 clock pulses:
  - T1: Starts the cycle. Generates control signals for latch, operation type, data direction, and address.
  - T2: Generates read/write control signals. DEN indicates output data is ready. READY indicates input data is ready.
  - T3: Reads/Writes data.
  - T4: Terminates control signals.

Auxiliary circuits

❖Bus Read Cycle



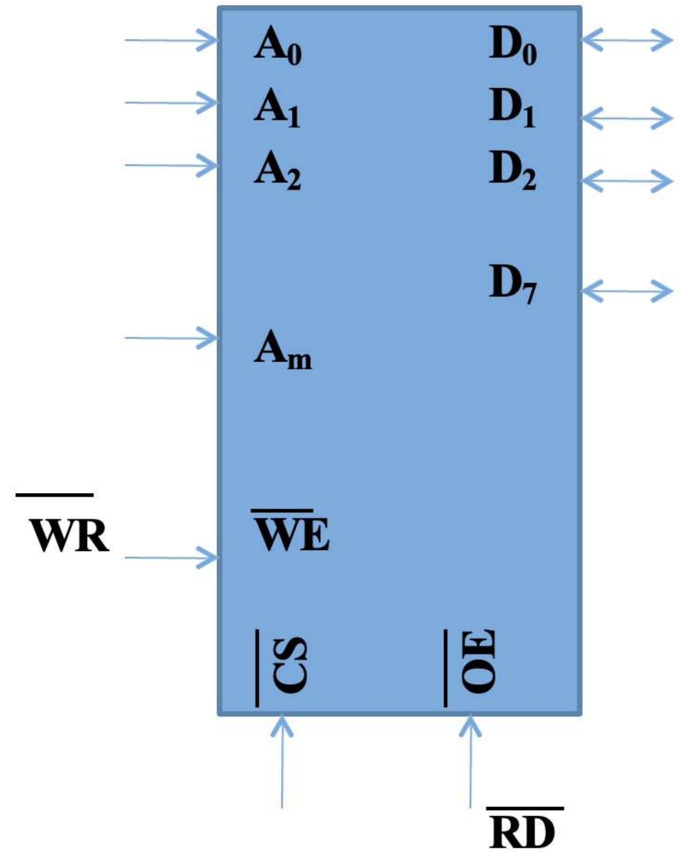
❖Bus Write Cycle



Introduction to Memory

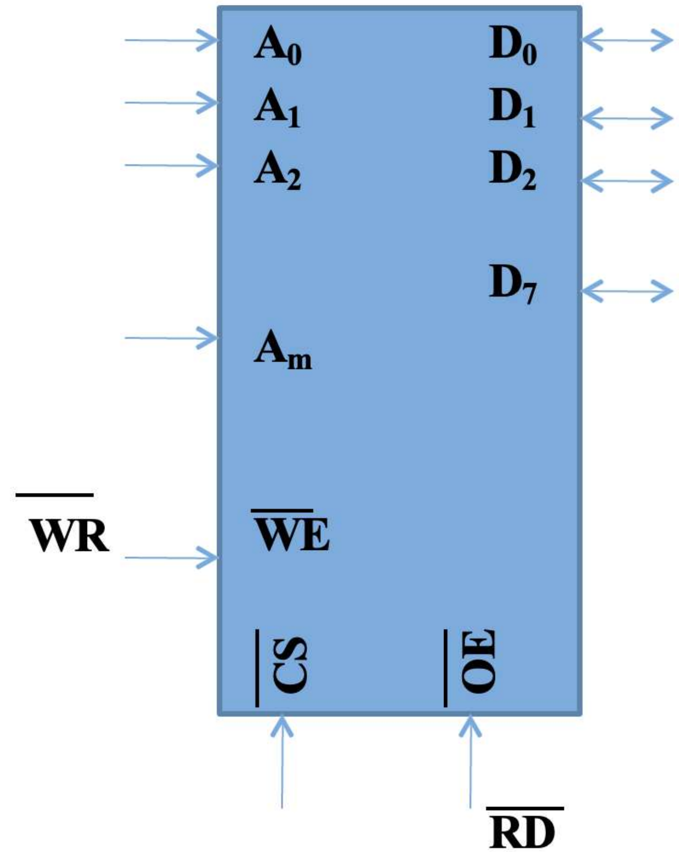
❖General Memory Circuit Structure:

- Address Signal Group:
  - The number of address lines corresponds to the memory circuit capacity.
  - m=20, memory circuit capacity is 220 = 1MB.
  - A0 - Am: 20 address line pins.



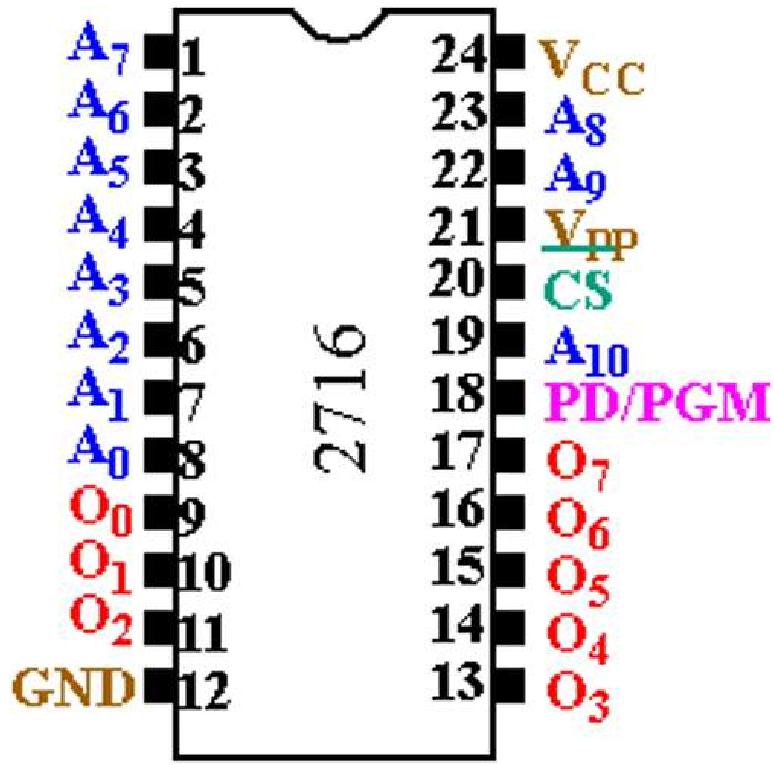
❖General Memory Circuit Structure:

- Data Signal Group:
  - The number of data lines determines the memory word length of the memory circuit
  - D0 - D7: 8 data pins.
- Typically, memory capacity and word length are stated simultaneously: 1Kx8 (capacity 1KB, word length 8 bits).
- Signals:
  - WE: Enable write,
  - OE: Enable output,
  - CS: Activate.



❖Intel 2176 (2Kx8) EFROM Memory Circuit:

- A0 - A10: Address signals
- 0 - O7: Data signals
- CS: Chip select (0 - read, 1 - write)
- PD/PGM: Maintain/Program Vpp = 25V
- A 2Kx8 memory chip occupies 2^{11} memory cells
- Therefore, 11 bits of internal chip address are needed (A0 - A10)



2K x 8 EPROM

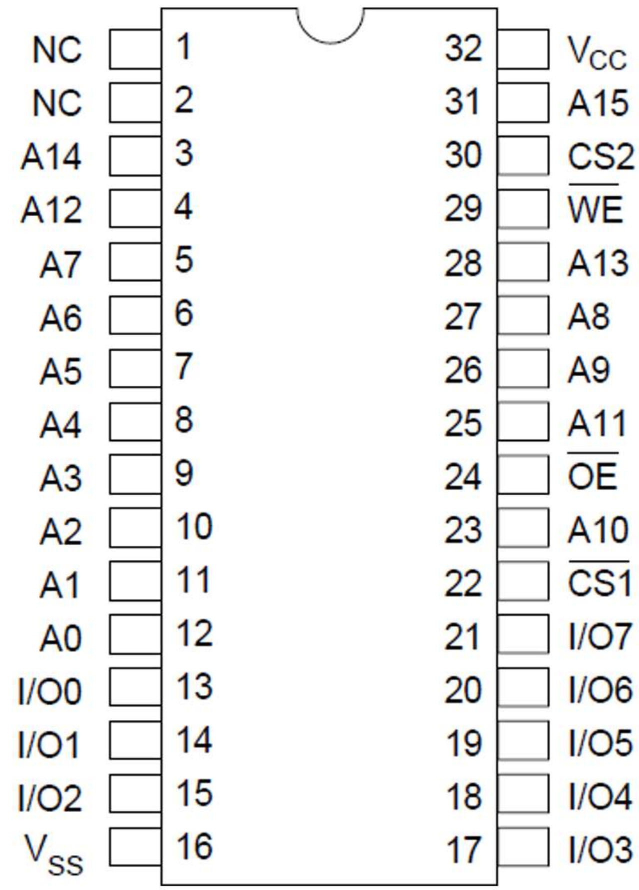
❖Memory circuit structure - SRAM

- A 64Kx8 memory chip occupies 216 memory cells
- Therefore, 16 bits of internal chip address are needed (A0 - A15)

Hitachi HM62864 - 64Kx8

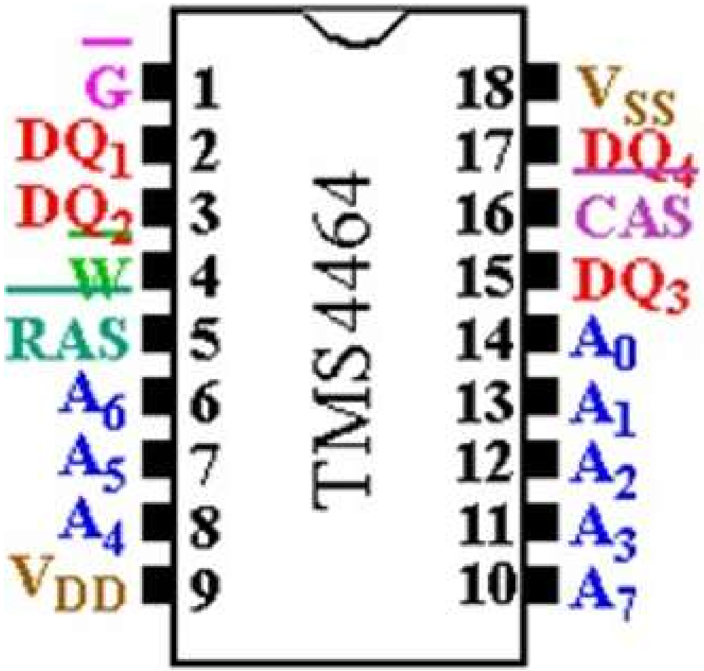
Tốc độ 50-85ns

Pin Name	Function
A0 to A15	Address
I/O0 to I/O7	Input/output
$\overline{CS1}$	Chip select 1
CS2	Chip select 2
$\overline{WE}$	Write enable
$\overline{OE}$	Output enable
NC	No connection
V_{CC}	Power supply
V_{SS}	Ground



❖ Memory circuit structure - TMS 4464 DRAM: 64Kx4

- 64K = {RA0 - RA7} +{CA0-CA7} ▶
- 64Kx4 memory chip occupies 216 memory cells
- Therefore, it requires 16 bits of internal chip address {RA0 - RA7} +{CA0-CA7} and the size of a memory word is 4 bits



64K x 4 DRAM

Pin(s)	Function
A_0-A_7	Address
DQ_1-DQ_4	Data In/Data Out
$\overline{RAS}$	Row Address Strobe
$\overline{CAS}$	Column Address Strobe
$\overline{G}$	Output Enable
$\overline{W}$	Write Enable

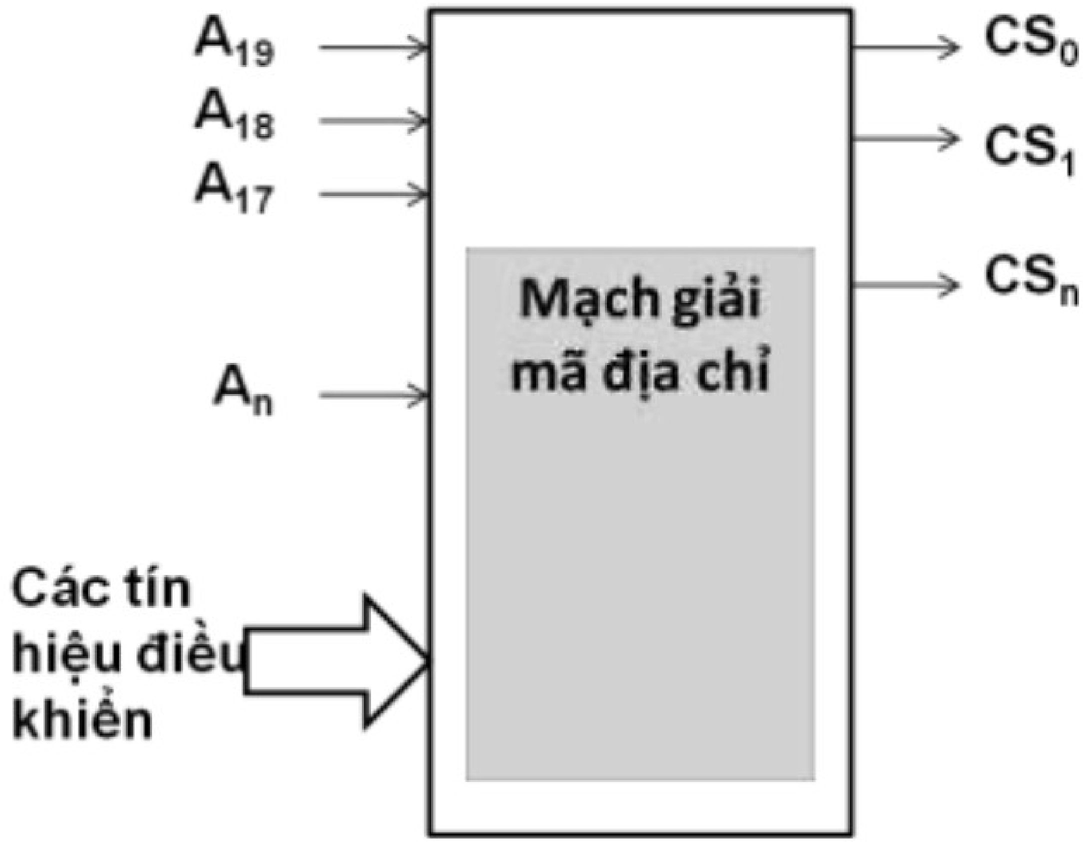
Matching CPU with memory

❖ Why is CPU-memory interfacing necessary?

- Each memory circuit interfacing with the CPU needs to be precisely referenced by the CPU when performing read/write operations.
- Each memory circuit is assigned a unique address space within the memory address space.
- Address assignment to a memory circuit is done via a microcontroller select pulse taken from the address decoder circuit.
  - Decoder circuit inputs:
    - 20-bit physical address signal
    - IO/M and RD (read) or WR (write) control signals
  - Types of memory circuits used:
    - ROM/EPROM
    - SRAM
    - DRAM

❖Interfacing circuits:

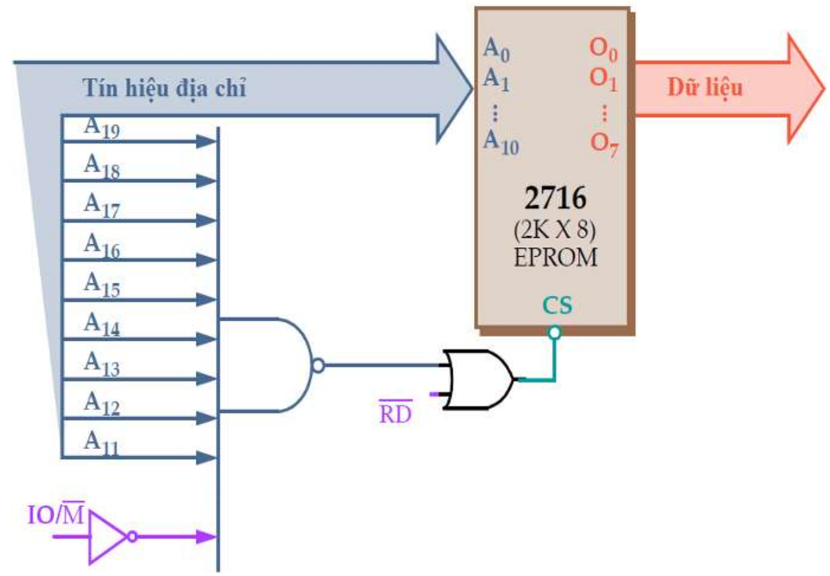
- Basic logic circuit: NAND
- Integrated circuits: 74LS139, 74LS138
- EPROM integrated circuit
- Address decoder circuit to select specific predefined memory modules



Matching CPU with memory

❖ Memory Address Decoder Circuit:

- Mapping address signals to memory chip select signals:
  - A19A18..An–> CS0, CS1, .., CSk
  - There are k+1 CSi memory chips
  - The decoder selects k+1 memory chips from the A19..An signals



- Full decoding:
  - Uses A19A18..An
  - The output signal selects only one memory chip.
- Reduced decoding:
  - Uses A19A18..Am; m > n outputs can select more than one memory chip.

Matching CPU with memory

❖ General Address Decoding Steps: Building an address decoding circuit for a ROM memory with capacity C_{IC}; Knowing that the size of a memory IC is IC and the base address is DCCS

- Step 1: Determine the number of bits for the internal chip address and the decoding circuit.
  - The number of bits for the internal chip is m = log2IC => the number of bits needed for the decoding circuit is 20 - m
- Step 2: Resolve the base address of the chips.
  - The number of memory chips is n = C_{IC}/IC => The capacity of one memory chip is IC.
  - A memory chip has: End address = Start address + Chip capacity - 1.
  - Address of IC 1: From DCCS to (DCCS + IC – 1H).
  - Address of IC 2: From (DCCS + IC) to (DCCS + 2xIC – 1H)
  - ...
- Step 3: Create a bit table
- Step 4: Draw the decoding circuit diagram

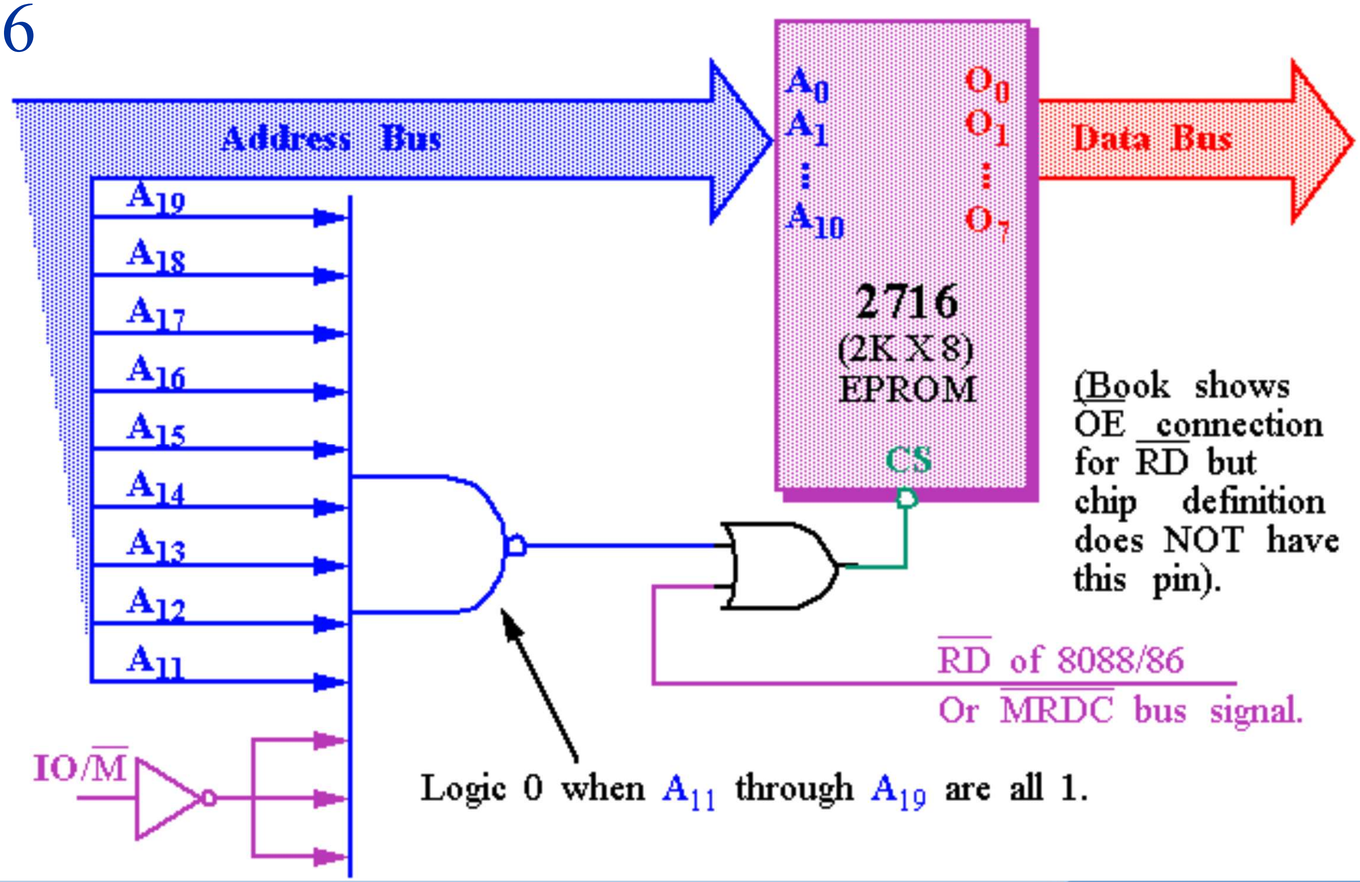
Matching CPU with memory

❖ Decoding memory addresses using basic logic circuits:

- Basic circuits include AND, OR, XOR, NOT, NAND, and NOR.
- Building a simple address decoder circuit with a limited number of outputs.
- Advantages:
  - Allows the creation of a complete decoder circuit.
  - Relatively simple and inexpensive when only one or a few outputs are needed.
- Disadvantages:
  - Bulky when decoding many outputs due to the rapidly increasing number of circuits.

Matching CPU with memory

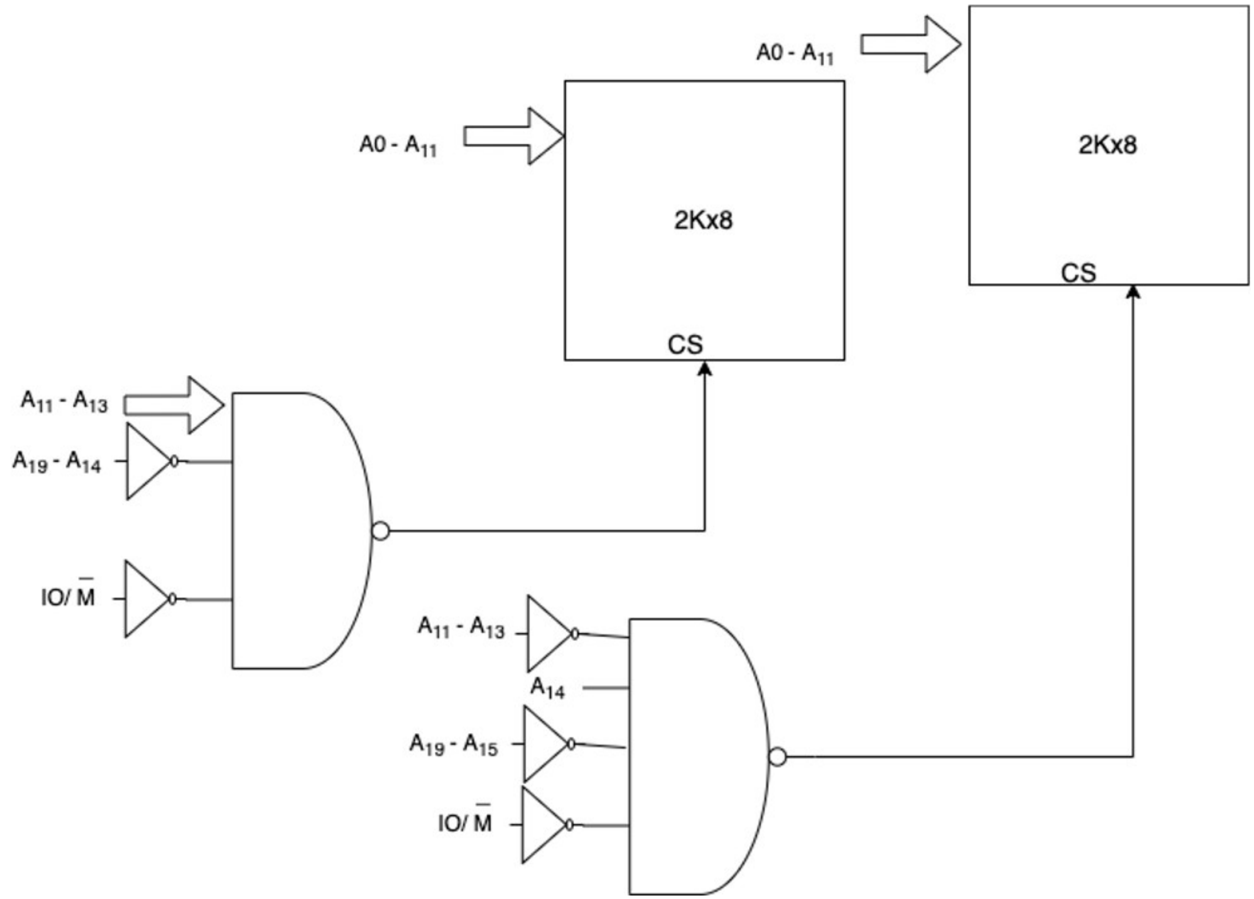
- Decoding memory addresses using basic logic circuits: Decoding memory addresses using basic logic circuits: Build an address decoder circuit for a 4KB ROM memory using basic logic circuits; knowing that the size of a memory IC is 2Kx8 and the base address is 03800H;
- Assume using 8086



❖ Decoding memory addresses using basic logic circuits:

C*ROM* = 4KB; IC = 2Kx8; ĐCCS = 03800H

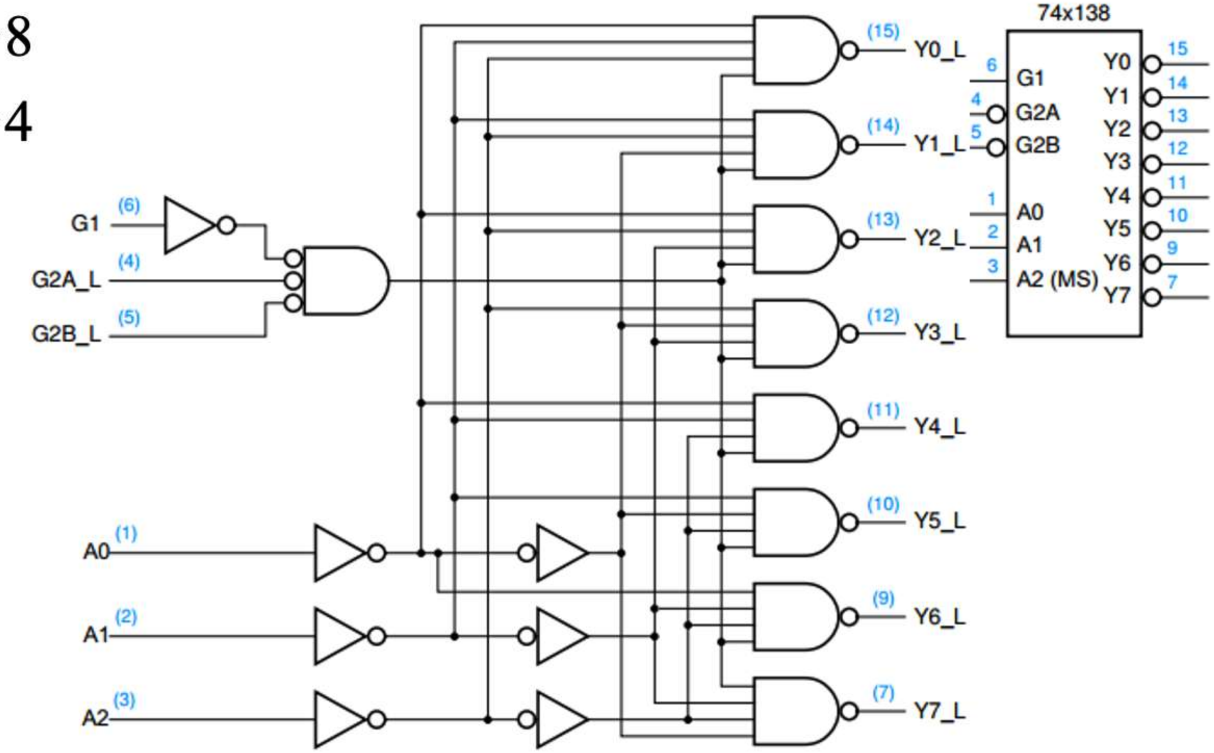
- Consider the 9 high bits (A11 – A19)
  - One for 03800H
  - One for 04000H



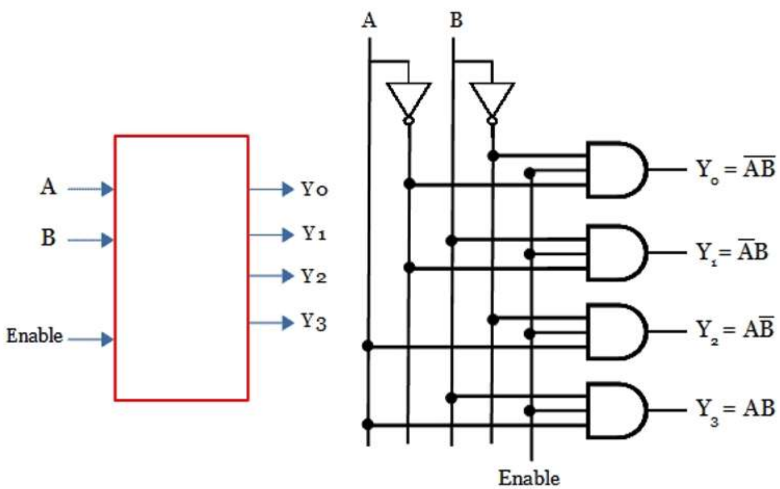
❖Memory address decoding using integrated circuits

- 74LS138 mạch giải mã 3→8
- 74LS139 mạch giải mã 2→4

74LS138



74LS139



Matching CPU with memory

❖ Memory Address Decoder using 74LS139 Integrated Circuit:

Construct an 8KB memory address decoder with a starting address of 0F800H using 2Kx8 memory chips. Use only 74LS139 address decoder chips (chips with 2 inputs and 4 outputs). CROM = 8KB; IC = 2Kx8; DCCS = 0F800H. Assume using 8086.

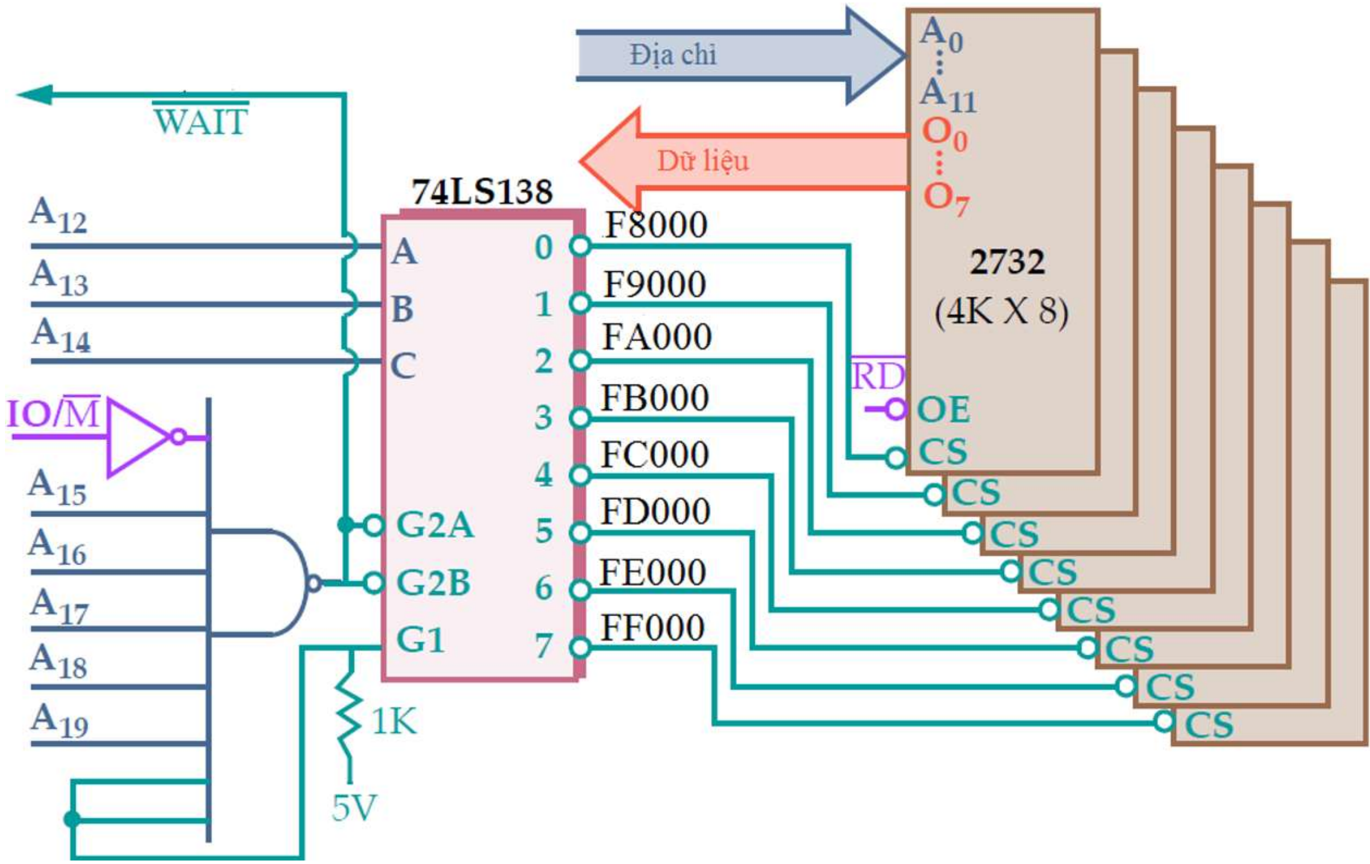
Matching CPU with memory

❖Memory Address Decoder using 74LS138 Integrated Circuit:

Construct an 8KB memory address decoder with a starting address of 0F800H using 2Kx8 memory chips. Use only 74LS138 address decoder chips (chips with 3 inputs and 8 outputs). CROM = 8KB; IC = 2Kx8; DCCS = 0F800H

❖ Decoding memory addresses using EPROM memory chips:

- 4Kx8 EPROM memory chip
- Address range: F8000 - FFFFFH (32KB), corresponding to 8 memory chips
- The number of internal signal lines in an EPROM memory chip is 12 bits (A0 - A11) for 4KB = 2^{12}.



Matching CPU with memory

❖ Decoding memory addresses using EPROM memory chips:

- The signal lines for decoding/chip selection are 20 - 12 = 8: A19...A16A15...A12
- The 8 4Kx8 EPROM memory chips have base addresses of F8000H, F9000H, ..., FE000H, FF000H respectively.
  - These chips have a base address of 5 constant signals A19A18A17A16A15, always equal to 1.
  - There are 3 different signals A14A13A12.
  - A14A13A12 are fed into inputs A, B, C of the decoder circuit, while the remaining address signals A19A18A17A16A15 and the IO/M control signals are connected to the control signals of the 74LS138 (G2A, G2B). The G1 signal is always at logic level 1. The outputs of the 74LS138 are connected sequentially to memory circuits corresponding to the pre-assigned address range.

Matching CPU with memory

❖ Decoding memory addresses using EPROM memory chips:

- Advantages:
  - Allows for the creation of a fully functional decoding circuit.
  - Allows for the creation of a decoding circuit that accepts a limited number of input signals and produces a limited number of output signal selections.
- Disadvantages:
  - Not suitable for decoding circuits that need to accept a large number of input signals and produce many output signals.
  - Requires the use of additional auxiliary logic circuits for the integrated circuit to allow for full decoding.

Matching CPU with memory

❖Memory Address Decoding using PROM Memory Chips:

- ROM/PROM memory can be used as a decoder because:
  - It accepts a group of address and control input signals.
  - It generates a group of output data signals; the state of these data signals depends on the value previously stored in ROM.
  - If the output data signals are mutually exclusive, they can be used as memory chip select signals.

Matching CPU with memory

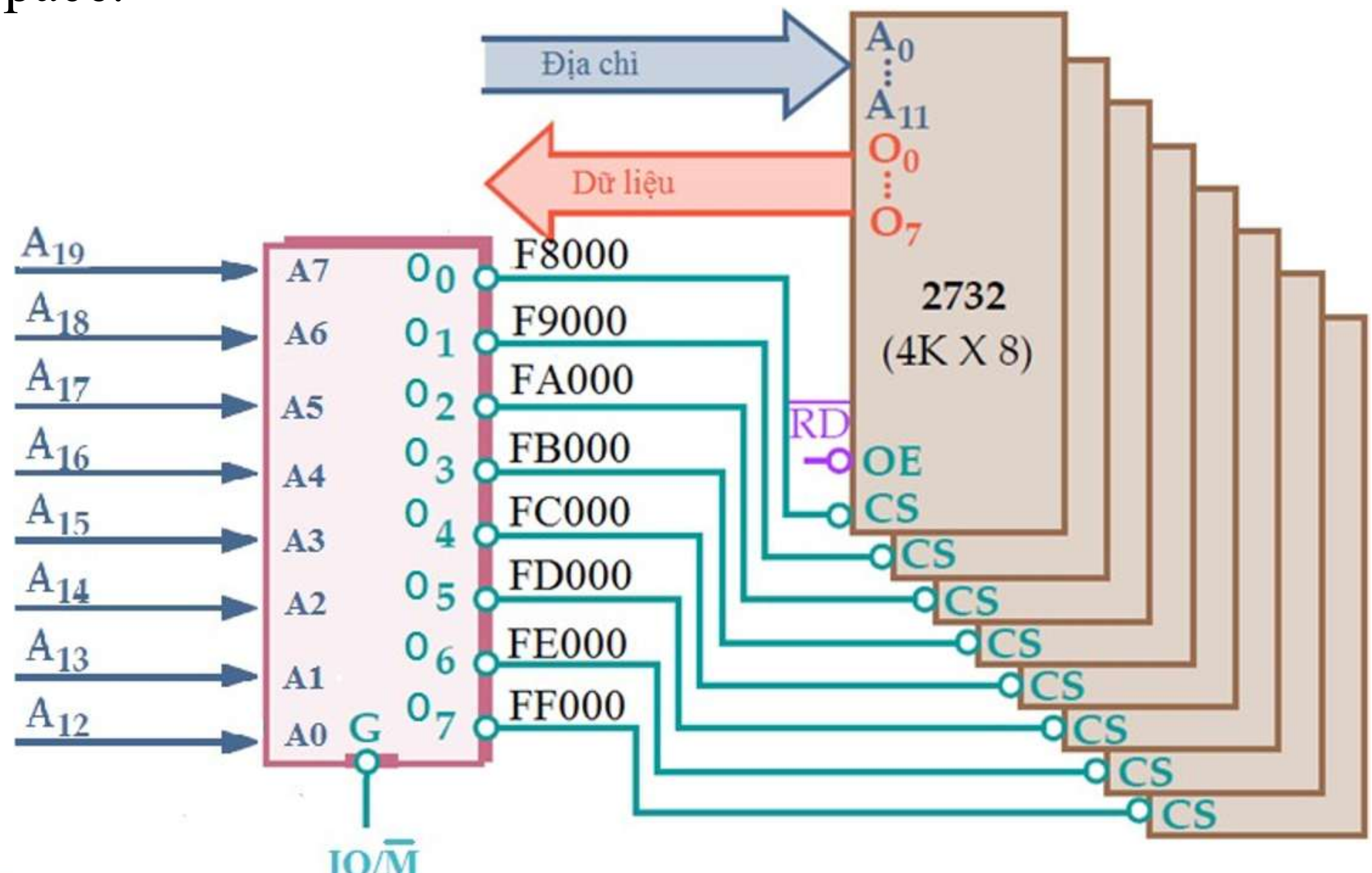
❖ Memory Address Decoding using PROM Memory Chips:

- Example: using a 256-byte PROM to decode 2732 (4Kx8) memory chips into the F8000-FFFFF address space.
  - Data pattern written to 256 bytes PROM:
    - Only 8 memory locations (on the diagonal) store low values (00)
    - All other memory locations store high values (FF)

A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	$\overline{O_0}$	$\overline{O_1}$	$\overline{O_2}$	$\overline{O_3}$	$\overline{O_4}$	$\overline{O_5}$	$\overline{O_6}$	$\overline{O_7}$
1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1
1	1	1	1	1	0	0	1	1	0	1	1	1	1	1	1
1	1	1	1	1	0	1	0	1	1	0	1	1	1	1	1
1	1	1	1	1	0	1	1	1	1	1	0	1	1	1	1
1	1	1	1	1	1	0	0	1	1	1	1	0	1	1	1
1	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1
1	1	1	1	1	1	1	0	1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0

❖ Memory Address Decoding using PROM Memory Chips:

- Example:
  - Use 256 byte PROM to decode 2732 4Kx8 memory chips into the F8000-FFFFF address space.

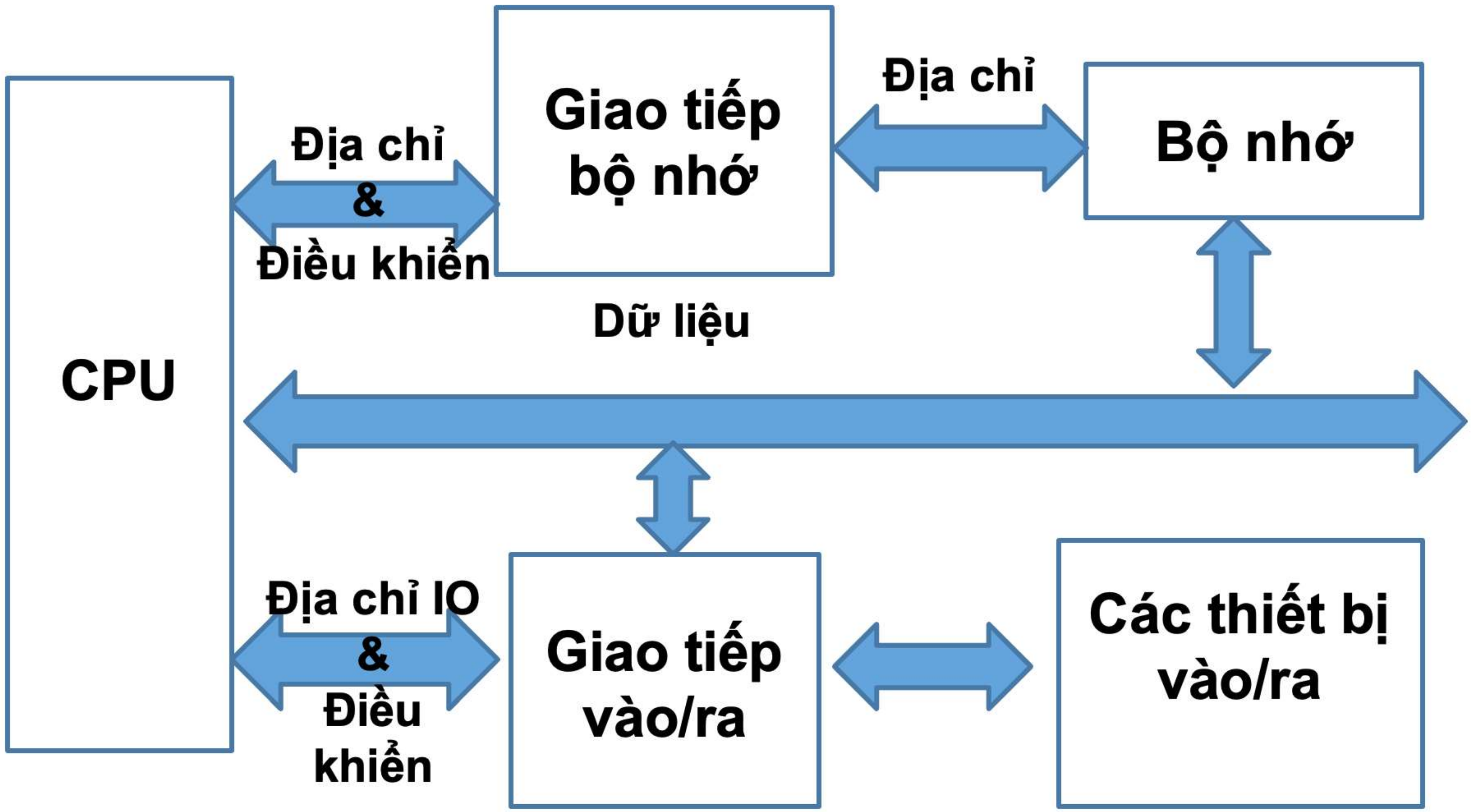


Matching CPU with memory

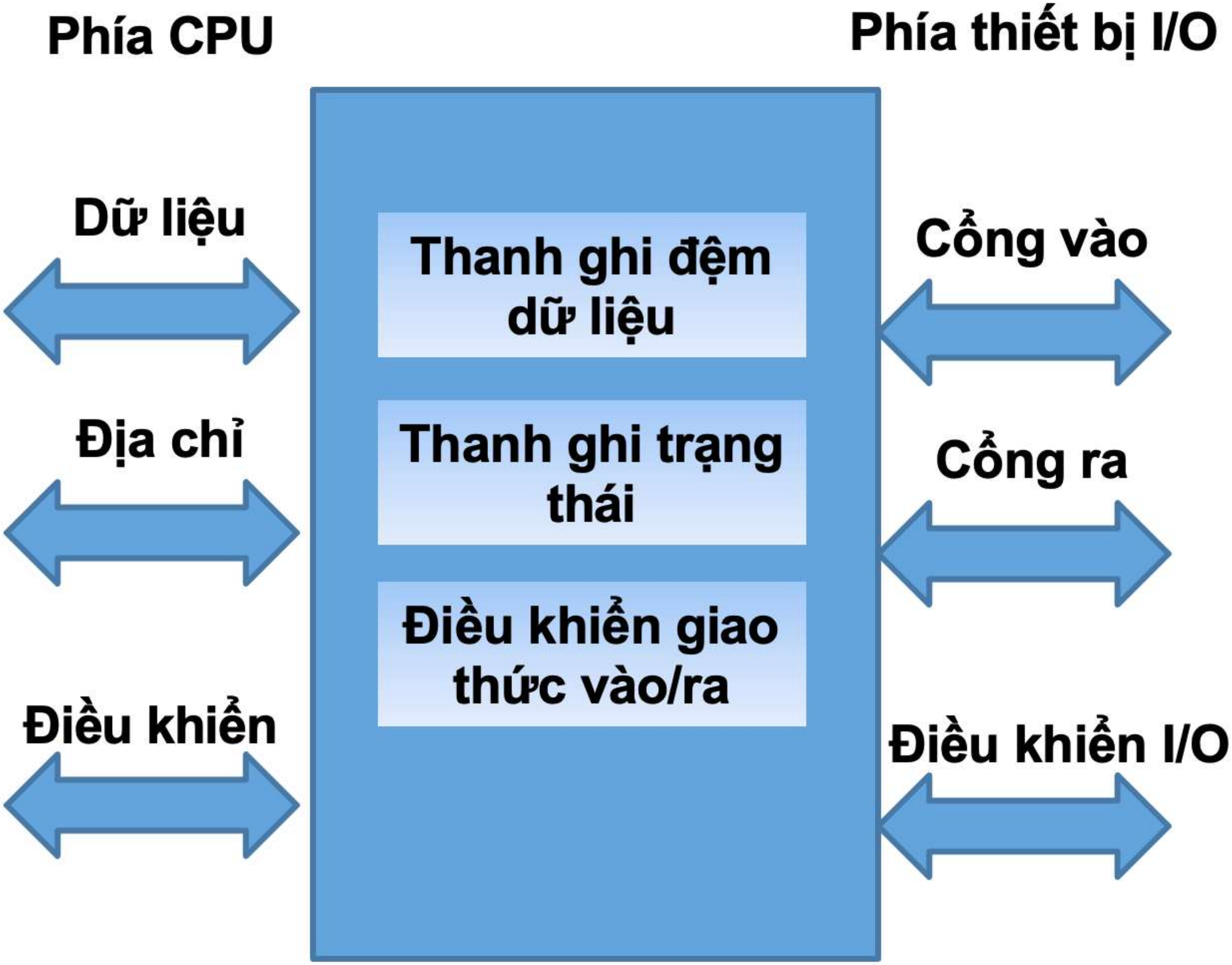
❖Memory Address Decoding using PROM Memory Chips:

- Advantages of address decoding using PROM memory chips:
  - Allows the creation of a complete decoder circuit without the need for auxiliary circuits ⇒ reduces decoder size.
  - Allows the creation of a decoder circuit that accepts multiple input signals and produces a large output circuit selection signal.
  - The state of the signals depends on the values stored in ROM; if these signals are mutually exclusive, they can be used as memory chip selection signals.
  - The addresses of the memory circuits can be easily changed by altering the location and data values within the ROM decoder memory circuit.

Pairing the CPU with input/output devices



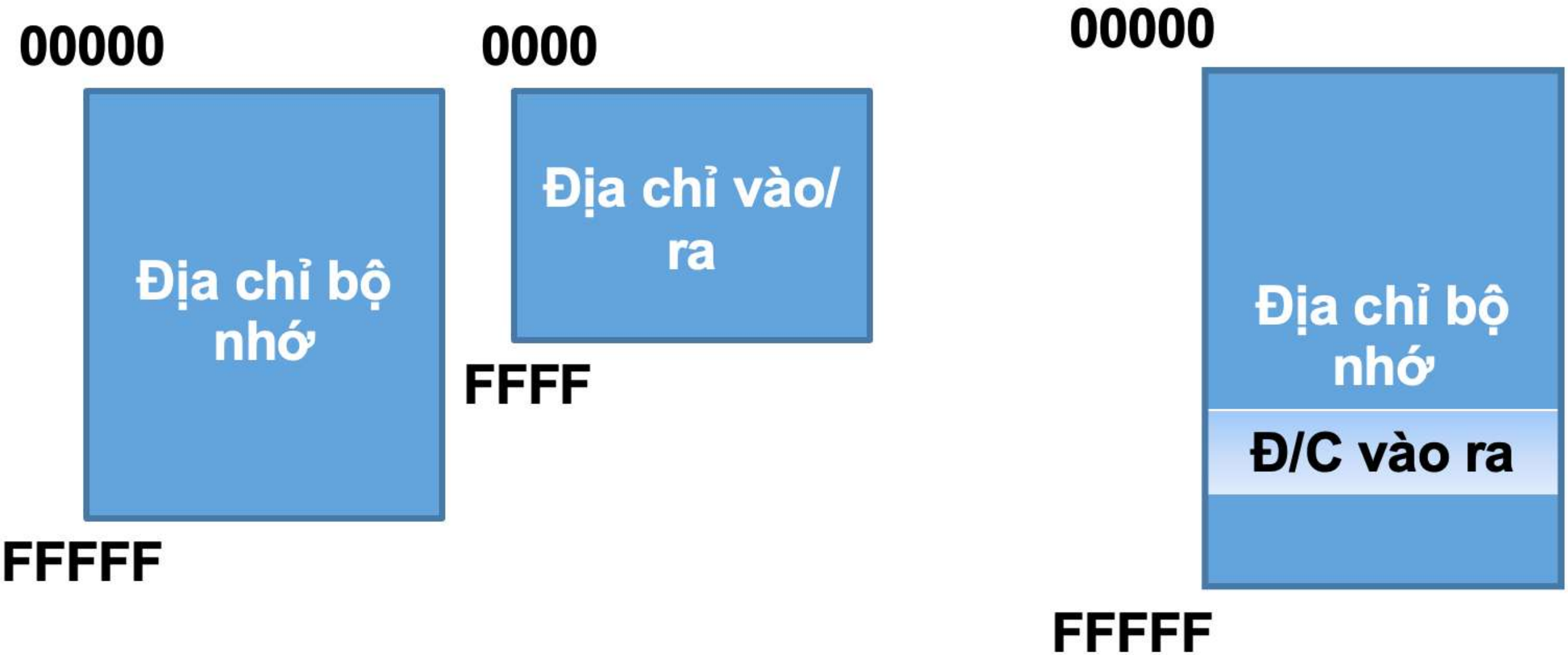
Pairing the CPU with input/output devices



Pairing the CPU with input/output devices

❖Classification of input/output devices by address space:

- Input/output devices with separate address spaces
- Input/output devices sharing an address space with memory



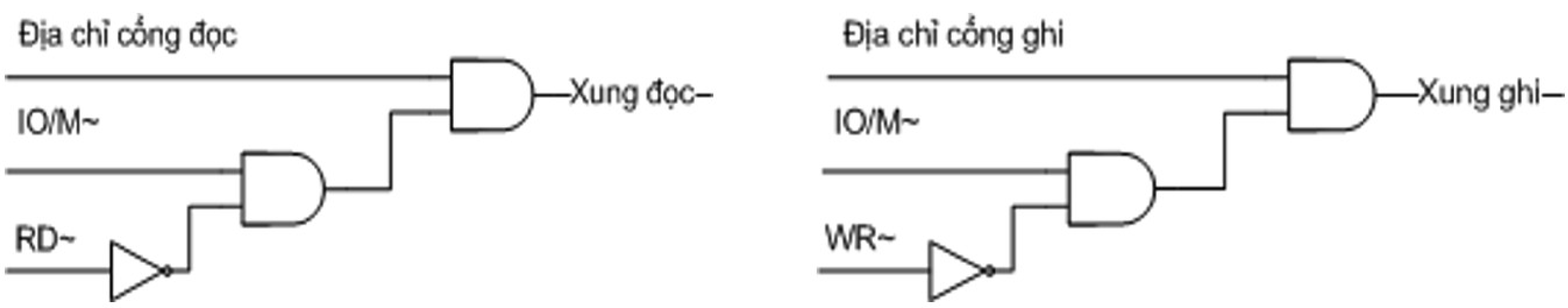
Pairing the CPU with input/output devices

❖ Classification of input/output devices by address space:

- Input/output devices with separate address spaces:
  - IN AX, [Port address]
  - OUT [Port address], AX
  - Input/output port address: 0000-FFFF is stored in DX; 00-FF is stored directly
- Input/output devices sharing an address space with memory:
  - MOV [Port address], AX
  - Read: MOV AX, [Port address]
  - Input/output port address 00000-FFFFF

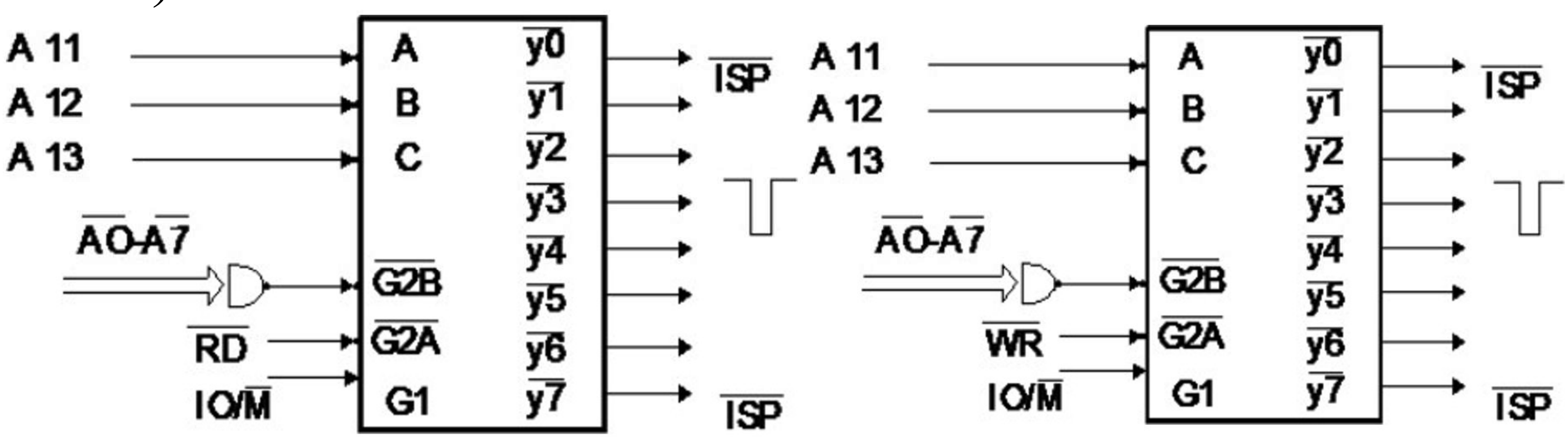
❖Decoding input/output device addresses using logic gates:

- Decoding input/output device addresses is similar to decoding memory addresses.
- Typically, the gates have 8-bit addresses (A0 - A7), in some microprocessors the gates may be 16 bits (A0 - A16).
- Combining address and control signals into read/write pulses.
- Simple decoding circuits are created using logic gates as follows:



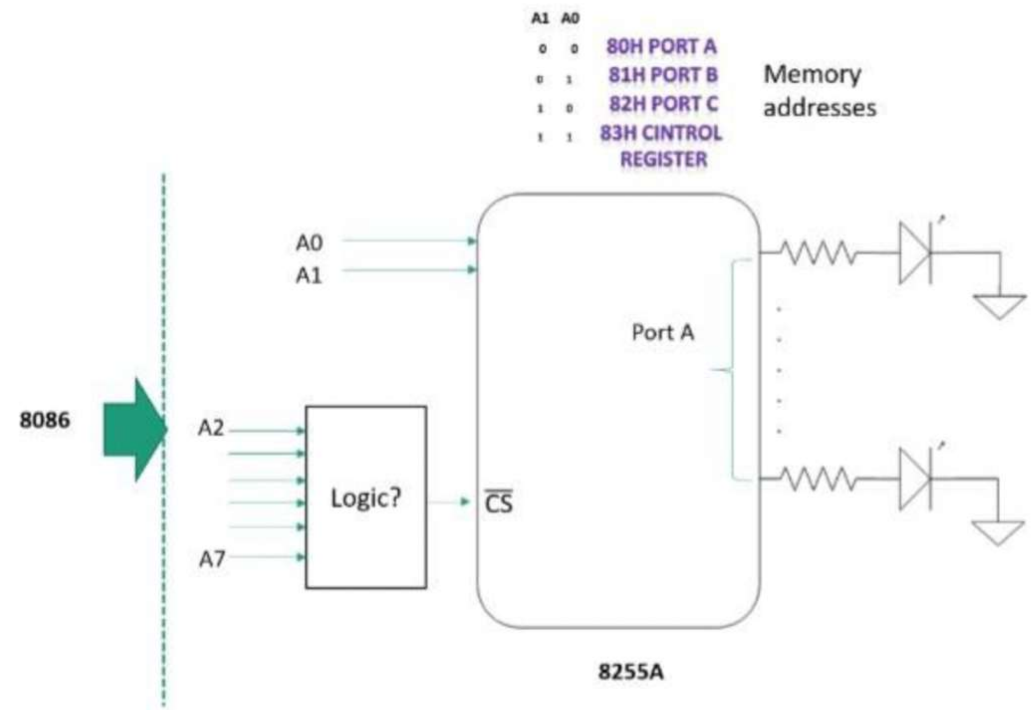
❖Decoding input/output device addresses using logic gates:

- Some simple gate circuits: Medium-sized integrated circuits can be used as interface gates with the microprocessor for data input/output. These circuits are usually composed of:
  - 8-bit latch circuits with 3-state output (74LS373, 74LS374)
  - 8-bit bidirectional buffer amplifier circuits with 3-state output (74LS245)

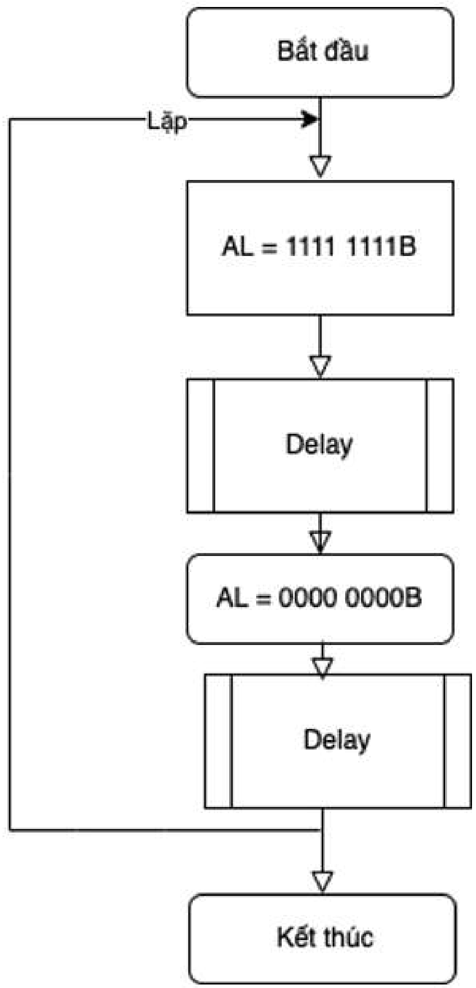


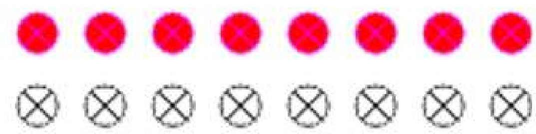
Giải mã địa chỉ cổng dùng 74LS138

❖**LED control program**: This program controls 8 LEDs connected to output port 2C0H, turning them on and off. The LEDs are turned on if the corresponding control bit receives a value of 0. Conversely, if the control bit is 1, the LEDs are off. All LEDs continuously switch on and off with a delay of 64 NOP command cycles between each switch.



❖LED control program:





Đặt giá trị cổng vào = FFH đèn tắt

Làm trễ 64 chu kỳ

Đặt giá trị cổng vào = 00H đèn sáng

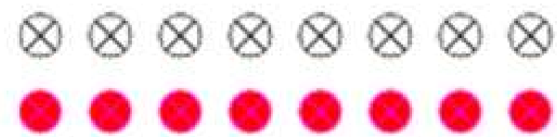
Làm trễ 64 chu kỳ

Some simple gate circuits

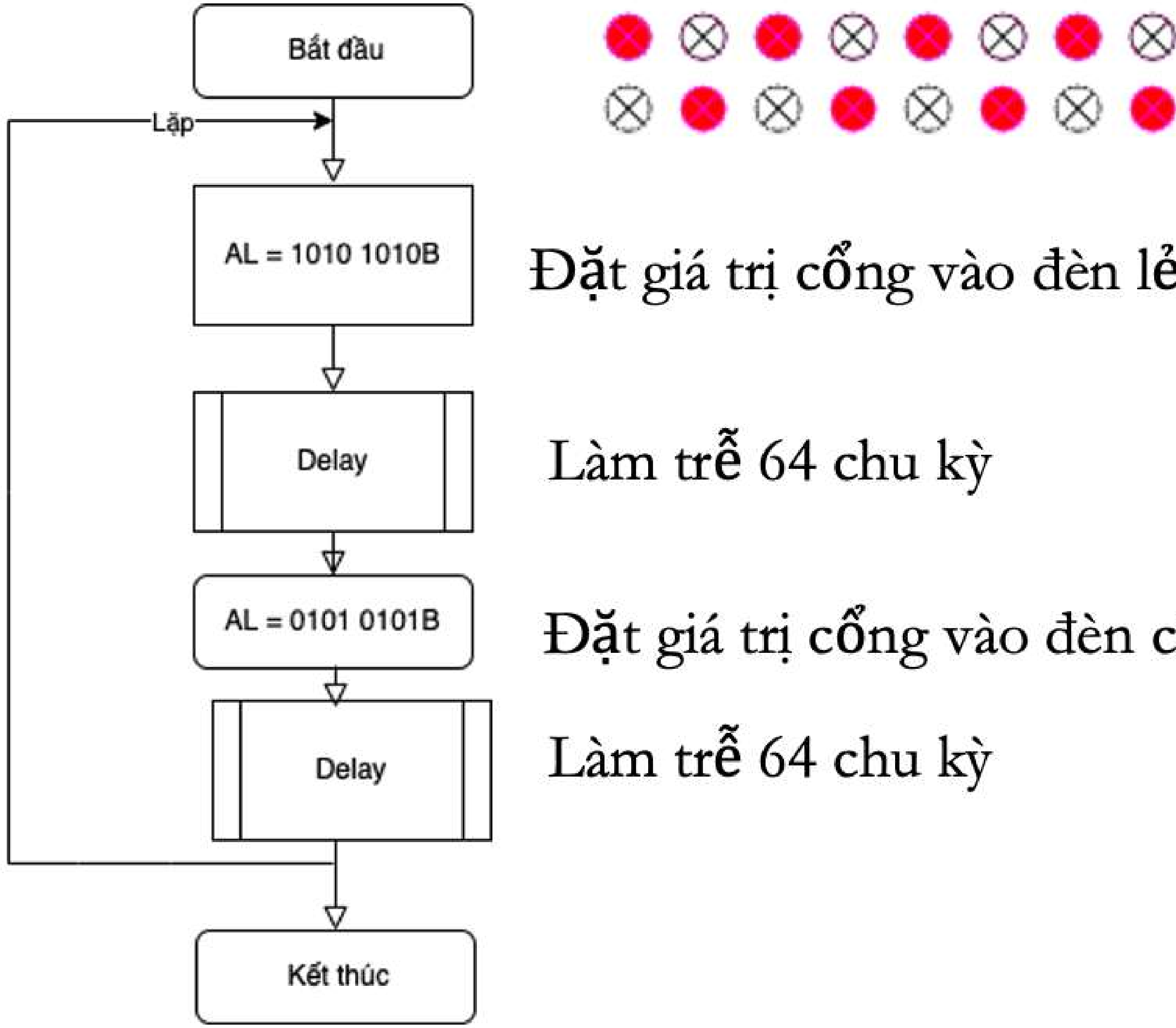
❖LED control program:

```
DK_LED EQU 2C0H   ; Cong dieu khien den LED

LAP: MOV AL, FFH ; Tat tat ca cac den LED
     MOV DX, DK_LED
     OUT DX, AL
     MOV CX, 64   ; Tre 64 chu ky NOP
TRE_1: NOP
    LOOP TRE_1
    XOR AL, AL    ; Dat AL = 0
    OUT DX, AL    ; Bat tat ca cac den
    MOV CX, 64
TRE_2: NOP
    LOOP TRE_2
    JMP LAP
```



❖LED control program:



Đặt giá trị cổng vào đèn lẻ sáng

Làm trễ 64 chu kỳ

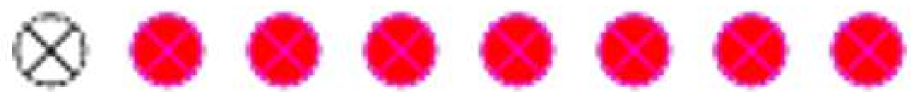
Đặt giá trị cổng vào đèn chẵn sáng

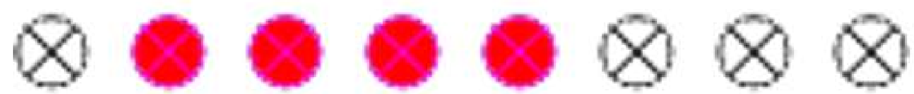
Làm trễ 64 chu kỳ

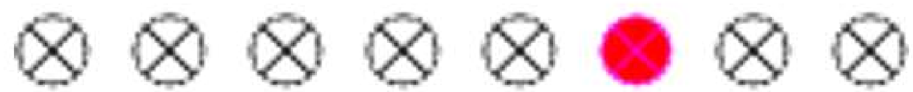
❖LED control program:

```
DK_LED EQU 2C0H   ; Cong dieu khien den LED
LAP: MOV AL, 0AAH
     MOV DX, DK_LED
     OUT DX, AL
     MOV CX, 64   ; Tre 64 chu ky NOP
TRE_1: NOP
    LOOP TRE_1
    MOV AL, 55H
    OUT DX, AL
    MOV CX, 64
TRE_2: NOP
    LOOP TRE_2
    JMP LAP
```

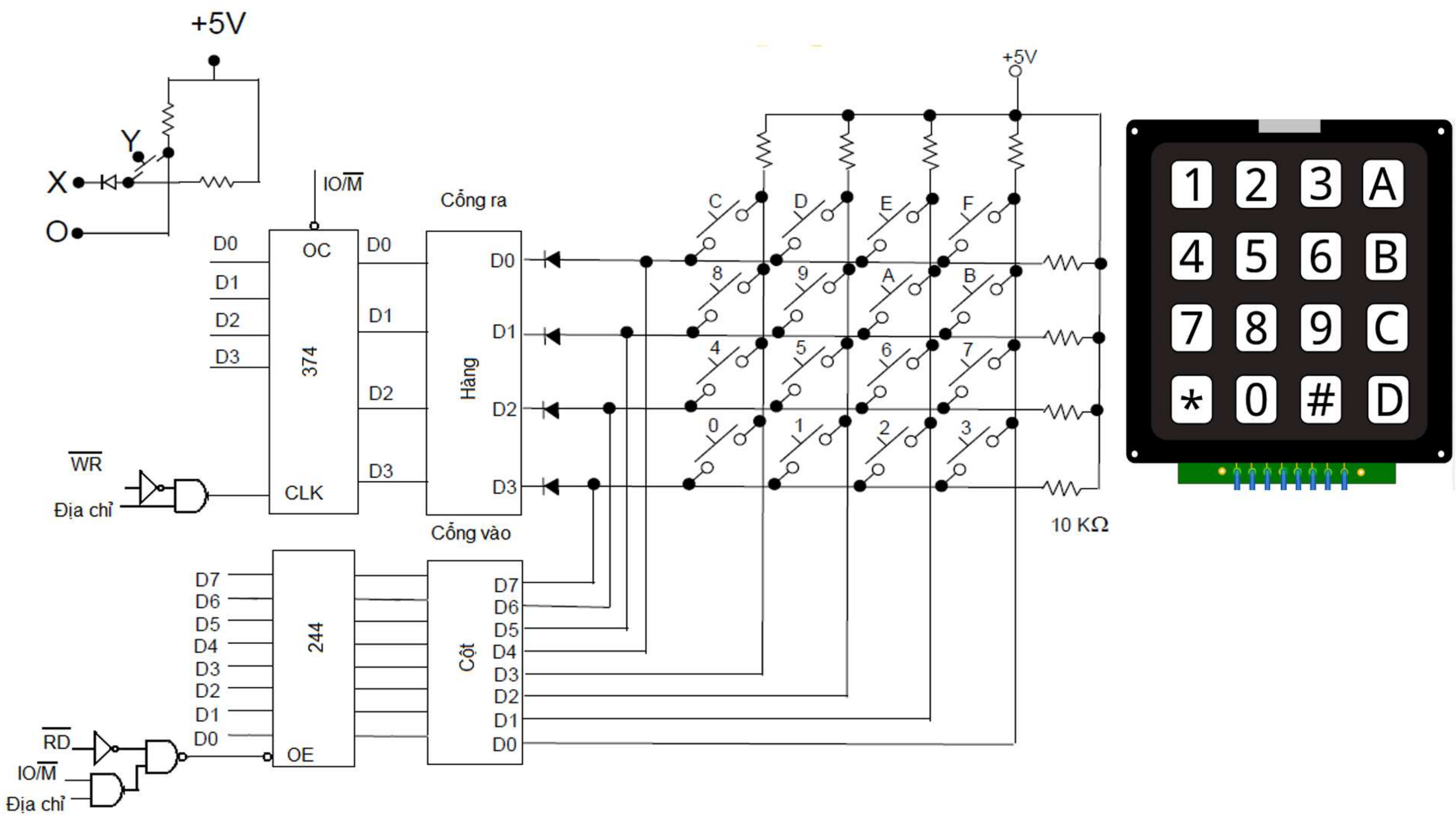
❖**LED control program**: Another exercises

1. **Đèn sáng tắt dần** 

2. **Sâu bò tới** 

3. **Sáng lần lượt** 

❖Keyboard Interfacing



❖Keyboard Interfacing

- 16-digit contact-type keyboard interface using integrated circuits:
  - 74LS374 IC is used to control row signals.
  - 74LS244 IC is used to control column signals.
- Operating principle:
  - When signal X is high (logic equals 1), the diode locks, and contact Y at terminal O receives a 5V voltage (no current).
  - When signal X is low (logic equals 0), the diode opens, and contact Y closes at terminal O, receiving a 0V voltage.
- Key press detection: sequentially scan rows and columns; if a key is pressed, the bits at output Di = 0.

❖Keyboard Interfacing

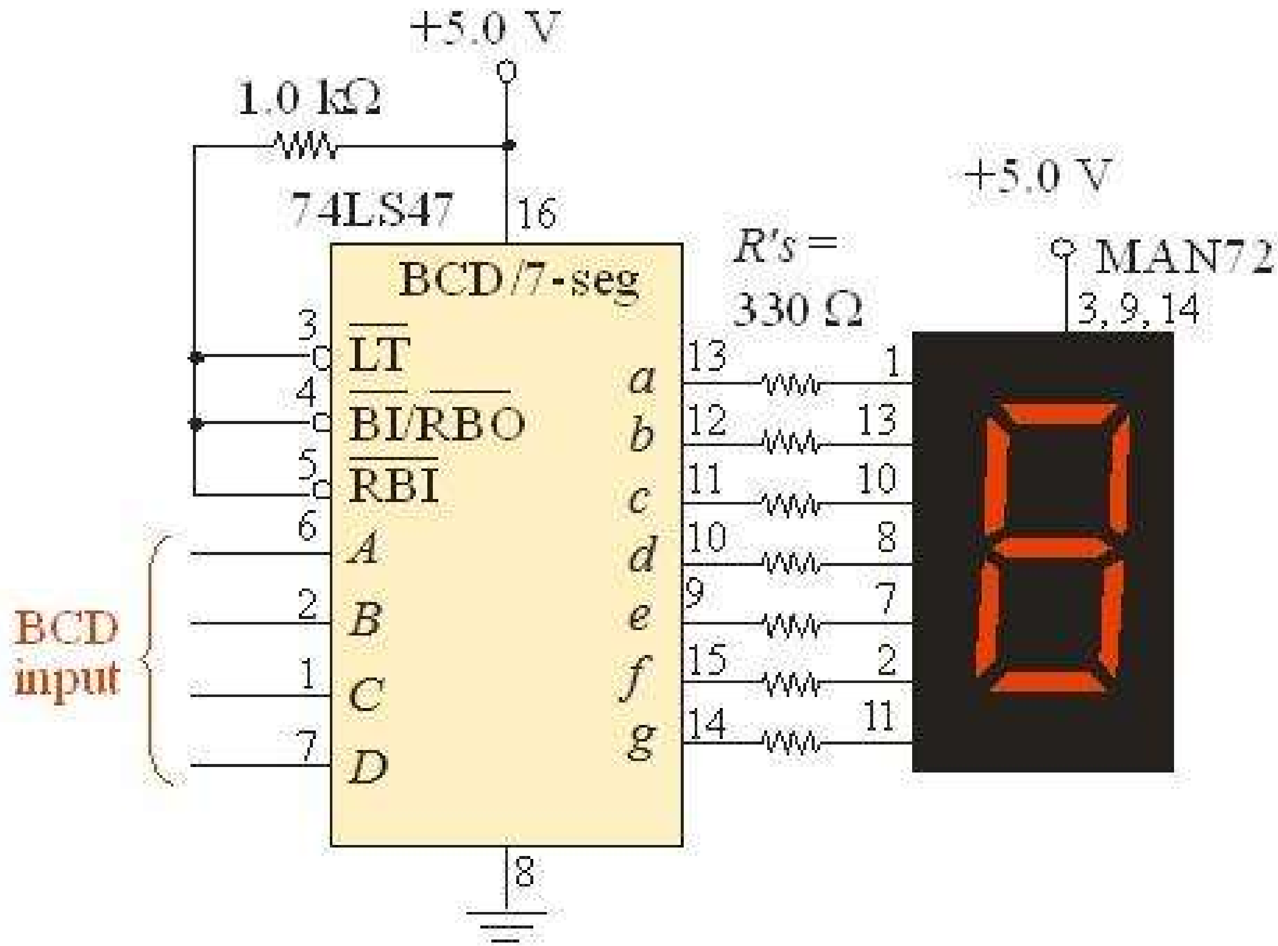
```
HANG EQU OAH    ; Dia chi Cong hang
COT EQU OBH    ; Dia chi Cong cot
     MOV AL, 1111 1110 ; Chi co DO = 0 - hang thu nhat
     OUT HANG, AL
KIEMTRA: IN AL, COT   ; Doc tin hieu cot
     AND AL, 0000 1000B ; Giu lai bit D3 tuong ung voi ban phim C
     JMP KIEMTRA    ; Khong bam
     ...     ; Phim C duoc bam
```

Some simple gate circuits

❖ Digital Display Interfacing

- Digital display interfacing using the 7447 integrated circuit and seven-segment LEDs:
  - Port A controls (turns on/off) the LED segments via 7 transistors Q1-Q7. Port A address is 0Ah.
  - Port B receives the digital data to be displayed - through the 7447 circuit, it decodes the input BCD data at port B (A-D) and generates activation signals (a-g) for the LED segments of the seven-segment LED. Port B address is 0Bh.
    - –Turn on the i-th LED: send bit Di = 0 to port A
    - –Turn off the i-th LED: send bit Di = 1 to port A
  - Display the number: Send the number to be displayed to port B, turn on the i-th LED by sending bit Di = 0 to port A.

❖Digital Display Interfacing



❖Digital Display Interfacing